

ANALOG INTEGRATED CIRCUITS LAB
AIC LAB MANUAL

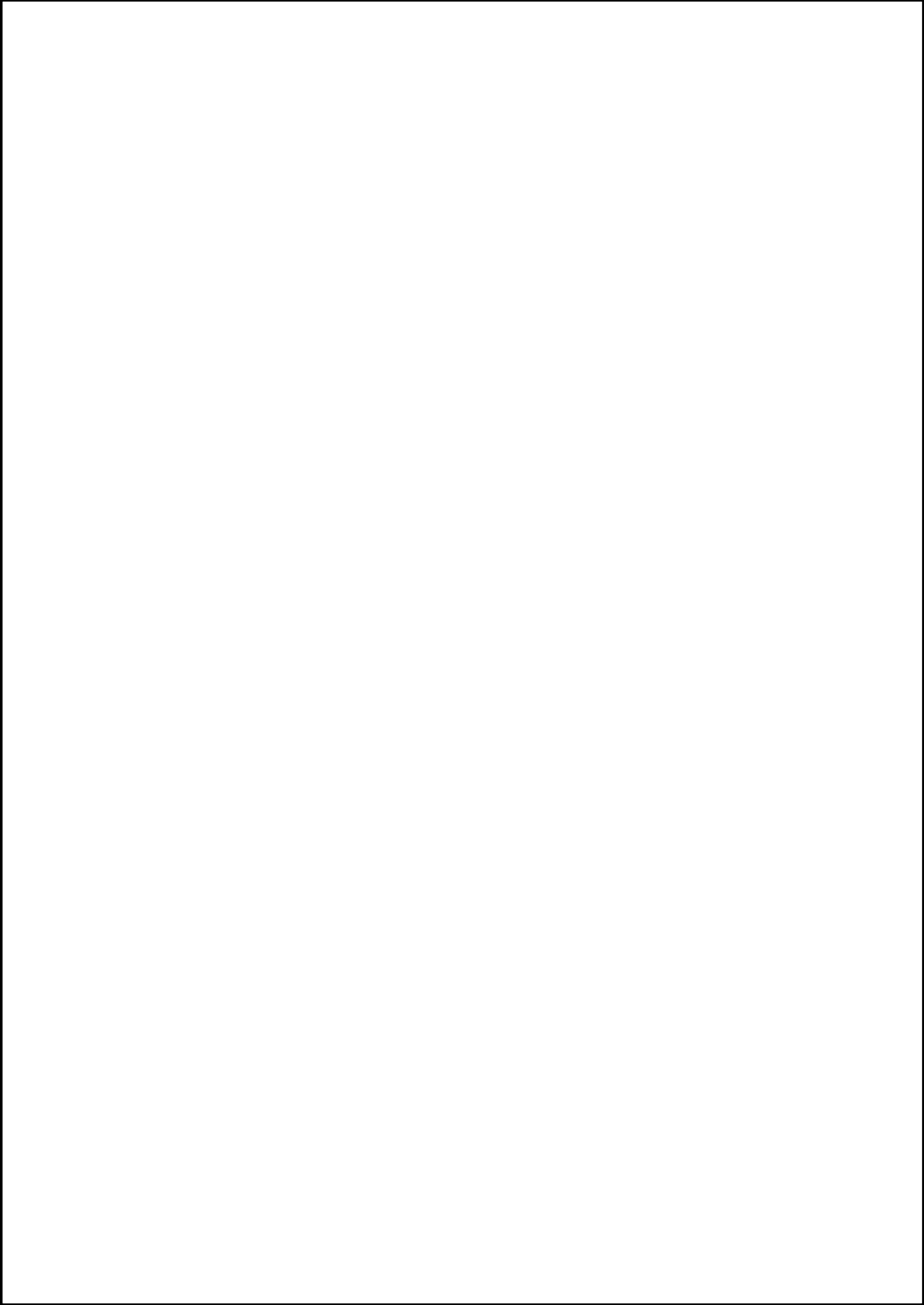
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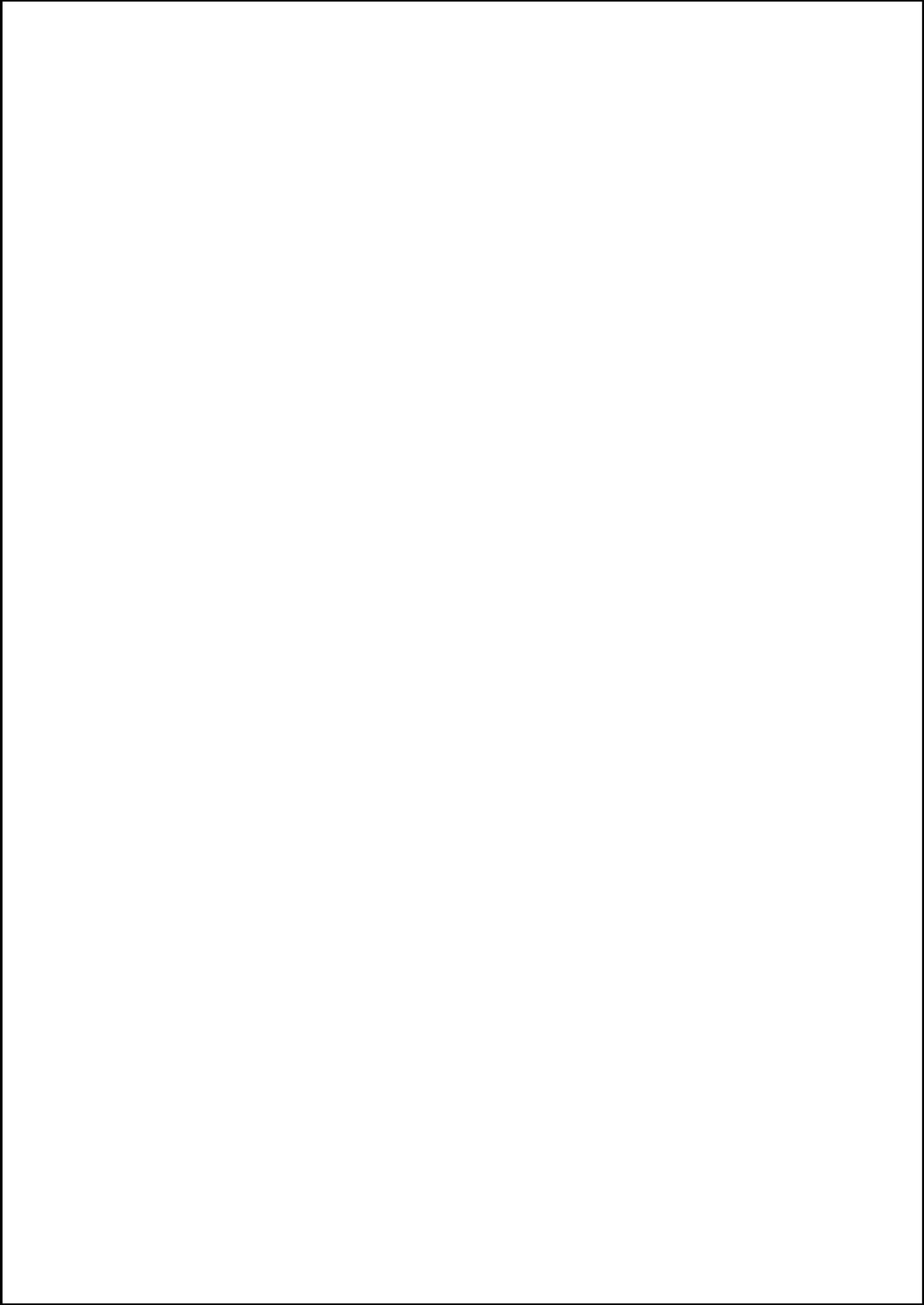
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July 16, 2026

List of Experiment

1. Inverting and Non-inverting Op-amp.
2. Applications of Op-amp (Adder, Subtractor, Adder-Subtractor, Average Amplifier).
3. Integrator and Differentiator using Op-amp.
4. Instrumentation Amplifier using Op-Amp.
5. Waveform generation using Op-Amp.
6. Design of Active filters (LPF, HPF) using Multisim/OrCAD/PSpice.
7. Simulation of Analog to Digital Converter.
8. Simulation of Digital to Analog Converter.
9. Simulation of Multivibrators.
10. Simulation of Wien Bridge and RC Phase Shift Oscillator.





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Ex No: 1	DESIGN AND IMPLEMENTATION OF SIGNAL CONDITIONING AMPLIFIERS USING OP-AMP
Date:	

Objective

To design and implement inverting and non-inverting amplifier circuits using IC 741 and analyze their characteristics, while understanding their roles in real-world applications such as sensor signal conditioning, audio electronics, and analog signal buffering.

Components /Equipments Required

S.No	Components	Range/Specification	Quantity
1	Op-Amp IC	741	1
2	Resistors	Refer design	As required
3	Function Generator	0-3MHz	1
4	Regulated Power Supply (RPS)	0 – 30V V	1
5	CRO	Dual Channel - 30 MHz	1
6	Breadboard and wires	–	As required

Theory

Inverting Amplifier

An inverting amplifier, the input signal is applied to the inverting terminal of the op-amp through a resistor R_i , while the non-inverting terminal is grounded. The output is fed back to the inverting terminal via a feedback resistor R_f . This configuration results in a 180° phase shift between the input and output signals.

- **Voltage Gain:**

$$A_v = -\frac{R_f}{R_i}$$

The negative sign indicates an inversion of phase.

- **Output Voltage:**

$$V_{\text{out}} = -\frac{R_f}{R_i} V_{\text{in}}$$

Non-Inverting Amplifier

In a non-inverting amplifier, the input signal is applied to the non-inverting terminal, while the feedback is connected from the output to the inverting terminal via a resistor network

Theory – Continued

consisting of R_f and R_i . The signal is amplified without inversion.

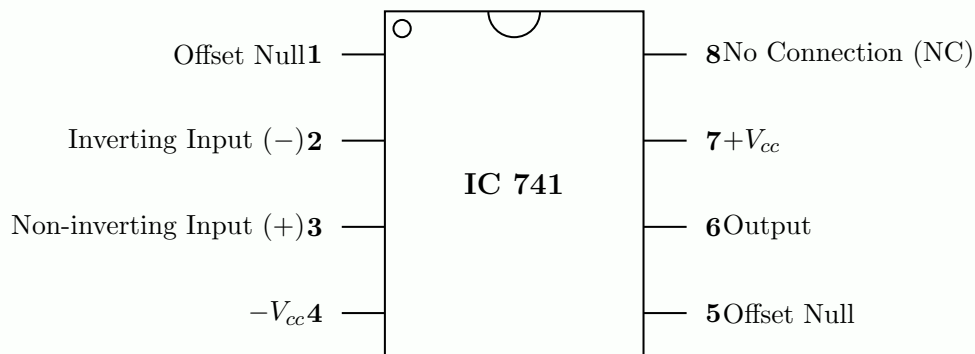
- **Voltage Gain:**

$$A_v = 1 + \frac{R_f}{R_i}$$

- **Output Voltage:**

$$V_{\text{out}} = \left(1 + \frac{R_f}{R_i}\right) V_{\text{in}}$$

Pin Configuration



Pin No.	Pin Name	Function
1	Offset Null	Used to nullify or correct the offset voltage error. Connected to a potentiometer between pin 1 and pin 5.
2	Inverting Input (-)	Input terminal where the signal is applied and inverted in phase at output.
3	Non-Inverting Input (+)	Input terminal where the signal is applied and remains in-phase at output.
4	V_{-} (Negative Supply)	Connected to negative voltage (e.g., -15V). Provides negative supply rail.
5	Offset Null	Used with pin 1 for offset voltage adjustment.
6	Output	Output terminal where the amplified signal appears.
7	V_{+} (Positive Supply)	Connected to positive voltage (e.g., $+15\text{V}$). Provides positive supply rail.
8	NC (No Connection)	Not internally connected; typically unused.

Design Calculation

1. Inverting Amplifier

- **Given Data:**

Input Voltage, $V_i = 1 \text{ V}_{\text{p-p}}$

Desired Gain, $A_v = -3$

Voltage gain of an inverting amplifier:

$$A_v = -\frac{R_f}{R_i}$$

Let: $R_i = 1.5 \text{ k}\Omega$; Calculate the feedback resistor R_f

$$R_f = |A_v| \times R_i = 3 \times 1.5 \text{ k}\Omega = 4.5 \text{ k}\Omega$$

Choose $R_f = 4.7 \text{ k}\Omega$ (nearest standard value)

- **Designed Values:**

$$R_i = 1.5 \text{ k}\Omega, \quad R_f = 4.7 \text{ k}\Omega$$

2. Non-Inverting Amplifier

- **Given:**

Input Voltage, $V_i = 1 \text{ V}_{\text{p-p}}$

Desired Gain value of Non Inverting amplifier is $A_v = 4$;

Voltage gain of a non-inverting amplifier:

$$A_v = 1 + \frac{R_f}{R_i}$$

Let assume the input resistance $R_i = 1.5 \text{ k}\Omega$

$$R_f = (A_v - 1) \times R_i = 3 \times 1.5 \text{ k}\Omega = 4.5 \text{ k}\Omega$$

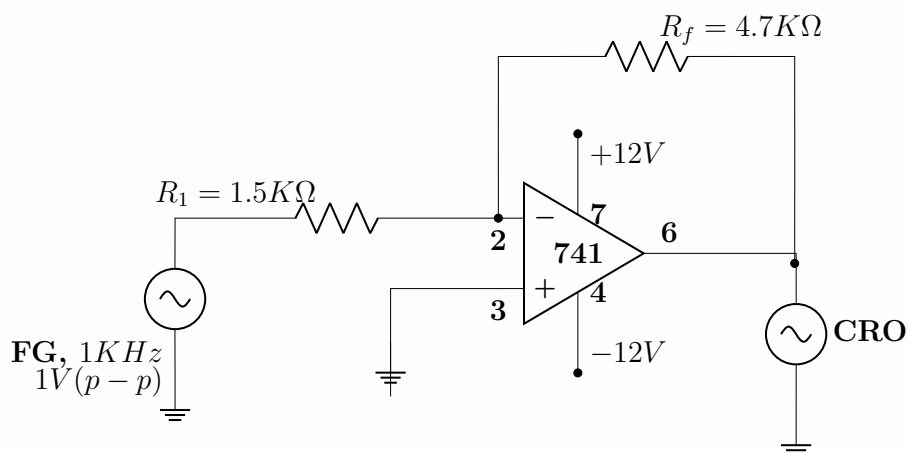
Choose the nearest standard resistor value $R_f = 4.7 \text{ k}\Omega$

Designed Values:

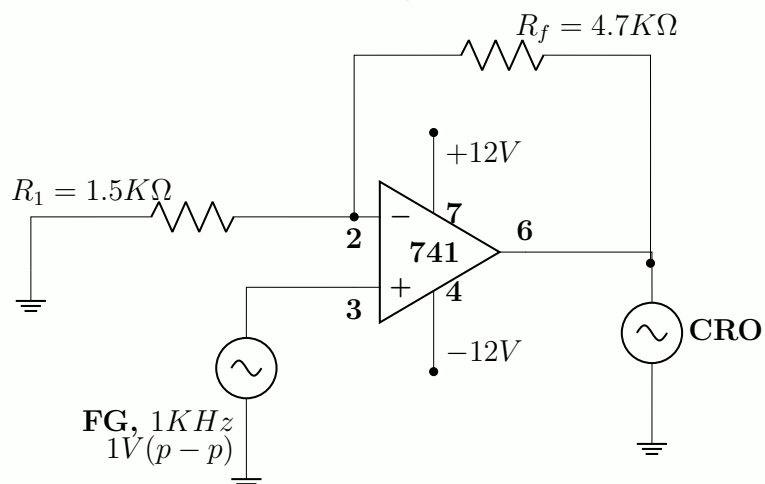
$$R_i = 1.5 \text{ k}\Omega, \quad R_f = 4.7 \text{ k}\Omega$$

Circuit Diagram

Inverting Amplifier

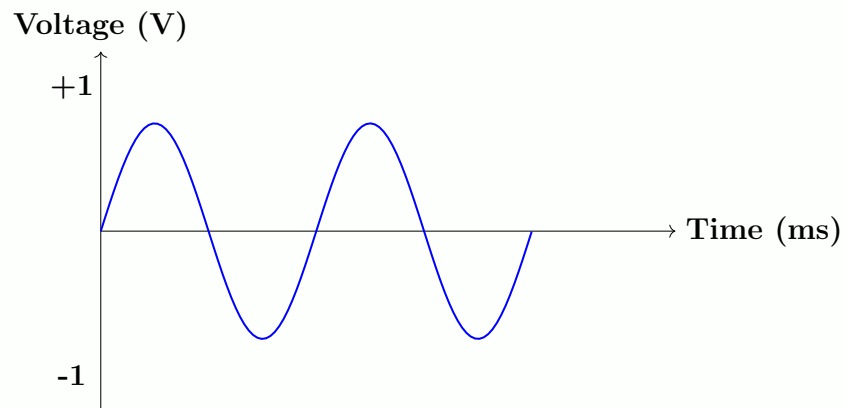


Non-Inverting Amplifier

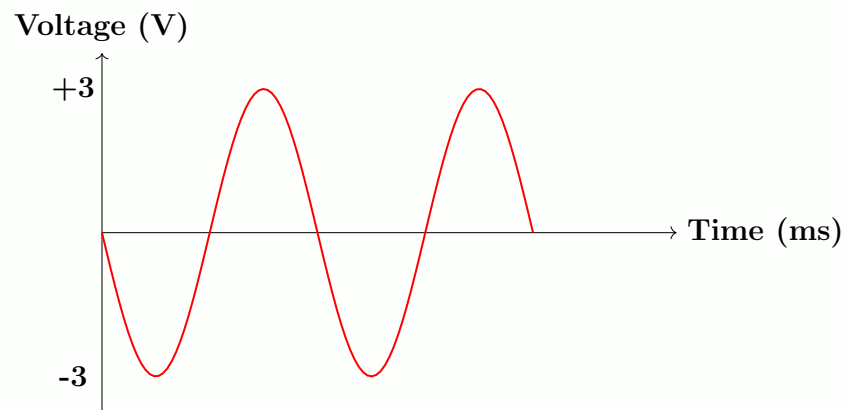


Model Graph

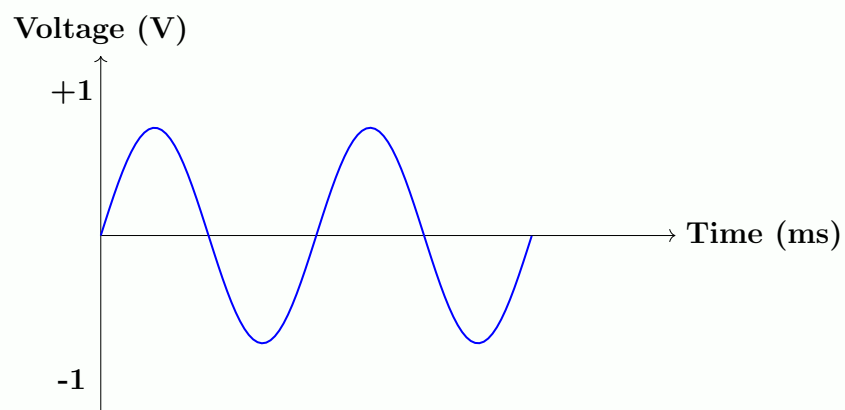
Inverting Amplifier - Input Signal

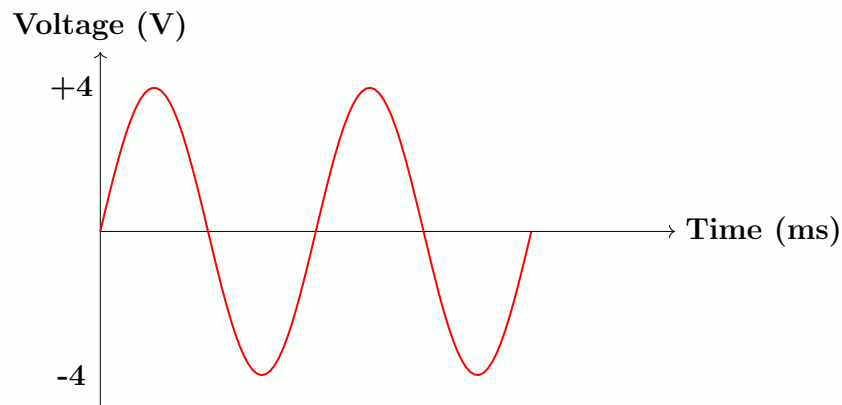


Inverting Amplifier - Output Signal (Inverted)



Non-Inverting Amplifier - Input Signal



Model Graph– Continued**Non-Inverting Amplifier - Output Signal (InPhase)****Procedure****1. Inverting Amplifier**

1. Connect the circuit as per the inverting amplifier configuration using IC 741.
2. Set a sinusoidal input signal of 1 V peak-to-peak (p-p) at 1 KHz using a function generator.
3. Observe both the input and output waveforms simultaneously using a Dual Trace CRO.
4. Tabulate the input and output voltages, calculate the gain, and verify it against theoretical values.
5. Plot the input and output waveforms on a graph sheet.

2. Non-Inverting Amplifier

1. Connect the circuit as per the non-inverting amplifier configuration.
2. Apply a 1 V (p-p) sinusoidal signal at 1 kHz to the non-inverting terminal using a function generator.
3. Use a Dual Trace CRO to observe both the input and output waveforms simultaneously.
4. Tabulate the measured input and output voltages, compute the gain, and compare with the theoretical result.
5. Plot the input and output waveforms on a graph sheet for analysis.

Tabulation

S.No.	Input voltage(V)	Inverting Amplifier			Non Inverting Amplifier		
	$V_{in}(V)$	$V_{out}(V)$	Gain (V/V)		$V_{out}(V)$	Gain (V/V)	
			T*	P *		T*	P*

*P - Practical value

*T -Theoretical value

Pre lab Viva Questions

1. What is the basic function of an operational amplifier (Op-Amp)?
2. What are the main differences between inverting and non-inverting amplifier configurations?
3. What is the formula for voltage gain in an inverting amplifier?
4. Why is the input resistance high in a non-inverting amplifier?
5. What is the phase relationship between the input and output in an inverting amplifier?

Post lab Viva Questions

1. If the input to an inverting amplifier is a 1 V peak sine wave and the gain is -5, what will be the output signal?
2. How would you select resistor values to design an inverting amplifier with a gain of -10?
3. How would you modify the non-inverting amplifier design to increase the voltage gain from 5 to 10 without changing the Op-Amp?
4. Given an op-amp circuit showing distortion at high frequencies, what could be a possible cause and how would you resolve it?
5. If you observe an inverted output in a non-inverting amplifier setup, what steps would you take to troubleshoot and correct the issue?

Result

The inverting and non-inverting amplifier circuits were designed and implemented using the IC 741 operational amplifier and also verified the gain values in both theoretical and practical value. These amplifier configurations are widely used in various practical applications such as:

- Analog signal conditioning
- Sensor interfacing systems
- Audio and communication electronics
- Impedance matching and buffering applications

Ex No:2	DESIGN AND REALIZATION OF ANALOG SIGNAL PROCESSING CIRCUITS USING OP-AMP
Date:	

Objective

To design and implement basic analog signal processing circuits such as adder, subtractor, adder-subtractor, and averaging amplifier using the IC 741 op-amp, and to understand their applications in sensor interfacing, audio mixing, and instrumentation.

Components /Equipments Required

Sl. No	Components	Range/Specification	Quantity
1	Operational Amplifier	IC 741	1
2	Resistors	Refer Design	As required
3	Bread Board	—	1
4	Regulated Power Supply	+12V, Dual	1
5	Multimeter	-	-
6	Connecting Wires	—	—

Theory

1. Inverting Adder (Summing Amplifier)

An inverting adder sums multiple input voltages applied to the inverting terminal through equal resistors. The non-inverting input is grounded, and the output is the negative sum of the inputs.

If V_1 , V_2 are two input signals given to the inverting terminal, then:

$$V_0 = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

If $R_1 = R_2 = R_f = R$, then:

$$V_0 = -(V_1 + V_2)$$

2. Subtractor

A subtractor circuit outputs the difference between two input voltages applied to both inverting and non-inverting terminals. When all resistors are equal, the output is the difference of inputs.

Theory – Continued

If V_1 and V_2 are input voltages at inverting and non-inverting terminals, then:

$$V_0 = \left(\frac{R_f}{R_2}\right) V_2 - \left(\frac{R_f}{R_1}\right) V_1$$

If $R_f = R_1 = R_2 = R$, then:

$$V_0 = (V_2 - V_1)V$$

3. Adder-Subtractor

This configuration allows simultaneous addition and subtraction of multiple inputs. Signals applied to the non-inverting terminal are added, and those to the inverting terminal are subtracted.

4. Average Amplifier

This circuit provides the average of all input voltages. Depending on the values of R_f and the input resistors R_a , R_b , R_c , it can act as a summing, scaling, or averaging amplifier.

For an ideal OPAMP, $v_1 = v_2$ and the current drawn by the OPAMP is zero. Applying KCL at node v_2 :

$$v_0 = -\frac{(v_a + v_b + v_c)}{3}V$$

when $R_a = R_b = R_c = R$ and $\frac{R_f}{R} = \frac{1}{3}$.

Applications Overview

- **Adder:** Used in audio mixing consoles to sum multiple audio signals.
- **Subtractor:** Applied in differential signal amplifiers to eliminate common noise in sensors.
- **Adder-Subtractor:** Utilized in signal conditioning circuits for adjusting offset and scaling of sensor outputs.
- **Averaging Amplifier:** Implemented in environmental data loggers for computing average sensor readings.

Design Calculation

Summing Amplifier

The circuit is based on an inverting summing amplifier configuration using an operational amplifier.

- Input voltages V_1 and V_2 are applied to the inverting terminal through resistors R_1 and R_2 .

Design – Continued

- The non-inverting terminal is grounded.
- A feedback resistor R_f connects the output to the inverting terminal.

Applying KCL at the inverting node

Assuming ideal op-amp operation,

$$V_- = V_+ = 0 \quad (\text{virtual ground})$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{-V_{\text{out}}}{R_f}$$

Multiplying through by R_f :

$$V_{\text{out}} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} \right)$$

Since $R_1 = R_2 = R_f = R = 10\text{ k}\Omega$, we get:

$$V_{\text{out}} = -(V_1 + V_2)$$

With equal resistors of $10\text{ k}\Omega$, the op-amp outputs the inverted sum of the inputs. This configuration is useful in analog summing applications such as audio mixers and weighted voltage addition circuits.

Subtractor

The circuit is a differential amplifier using an op-amp. It subtracts two input voltages: V_1 (inverting input) and V_2 (non-inverting input).

Voltage at Non-Inverting Terminal

Using the voltage divider rule:

$$v_+ = V_2 \cdot \frac{R_3}{R_2 + R_3}$$

Since $R_2 = R_3$, we get:

$$v_+ = \frac{V_2}{2}$$

Design – Continued**Applying KCL at Inverting Terminal**

Let the voltage at the inverting terminal be v_- . For an ideal op-amp, $v_- = v_+ = \frac{V_2}{2}$

$$\frac{V_1 - v_-}{R_1} = \frac{v_- - V_{\text{out}}}{R_f}$$

Assuming $R_1 = R_f = R$, and substituting $v_- = \frac{V_2}{2}$:

$$V_1 - \frac{V_2}{2} = \frac{V_2}{2} - V_{\text{out}}$$

$$V_1 - V_2 = -V_{\text{out}} \Rightarrow V_{\text{out}} = V_2 - V_1$$

By choosing all resistors equal to $10\text{ k}\Omega$, the circuit outputs the difference between two input voltages. This configuration is widely used in instrumentation and sensor signal processing.

Average Amplifier

Calculate the required feedback resistor (R_f) for an averaging amplifier using an operational amplifier with three inputs, each having a resistance of $R = 10\text{ k}\Omega$.

Given

- Number of inputs, $n = 3$
- Input resistors: $R_a = R_b = R_c = R = 10\text{ k}\Omega$
- The desired output voltage:

$$V_o = -\frac{V_a + V_b + V_c}{3}$$

For an inverting summing amplifier:

$$V_o = -\left(\frac{R_f}{R}\right)(V_a + V_b + V_c)$$

To obtain the averaging behavior:

$$\frac{R_f}{R} = \frac{1}{3}$$

Calculation

$$R_f = \frac{R}{3} = \frac{10\text{ k}\Omega}{3} = 3.33\text{ k}\Omega$$

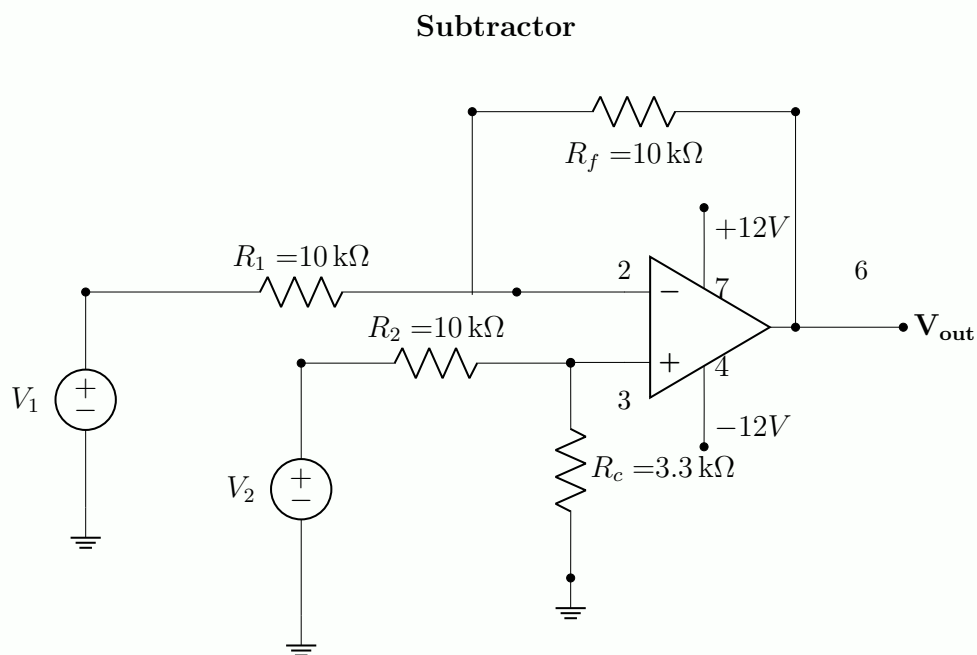
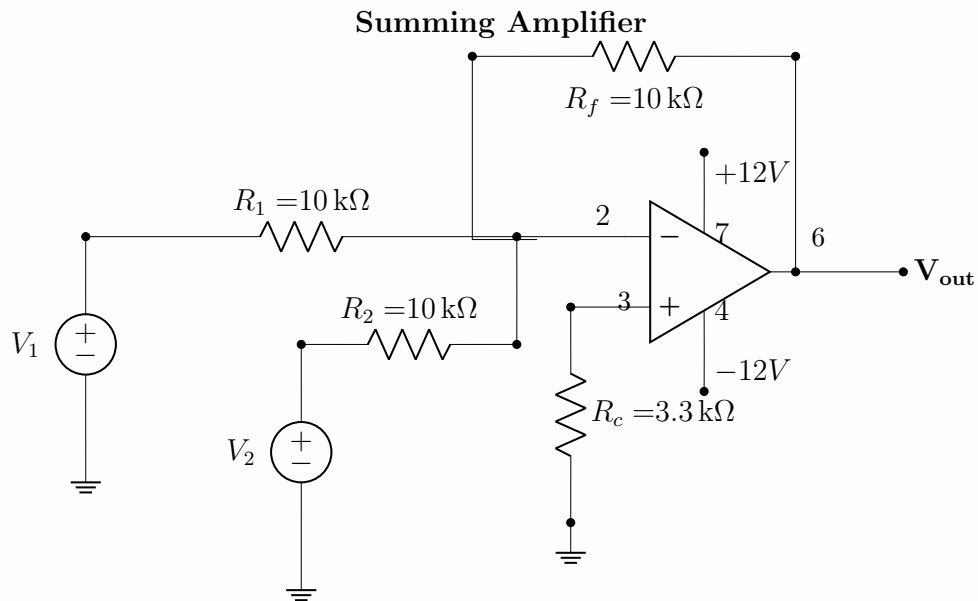
Design – Continued

The feedback resistor required is:

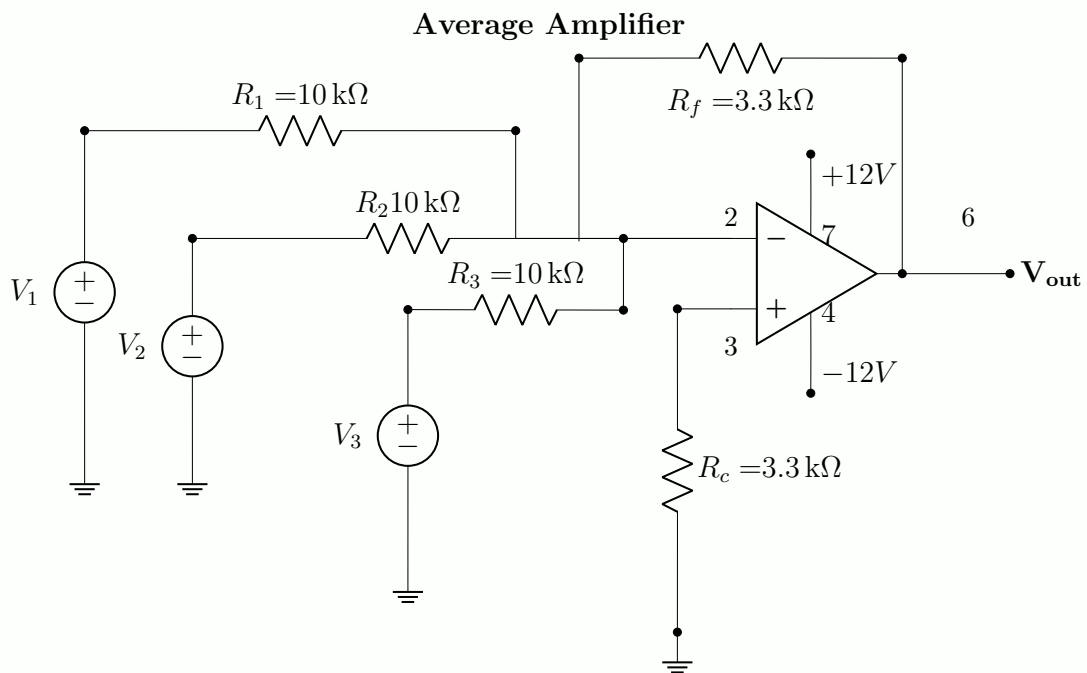
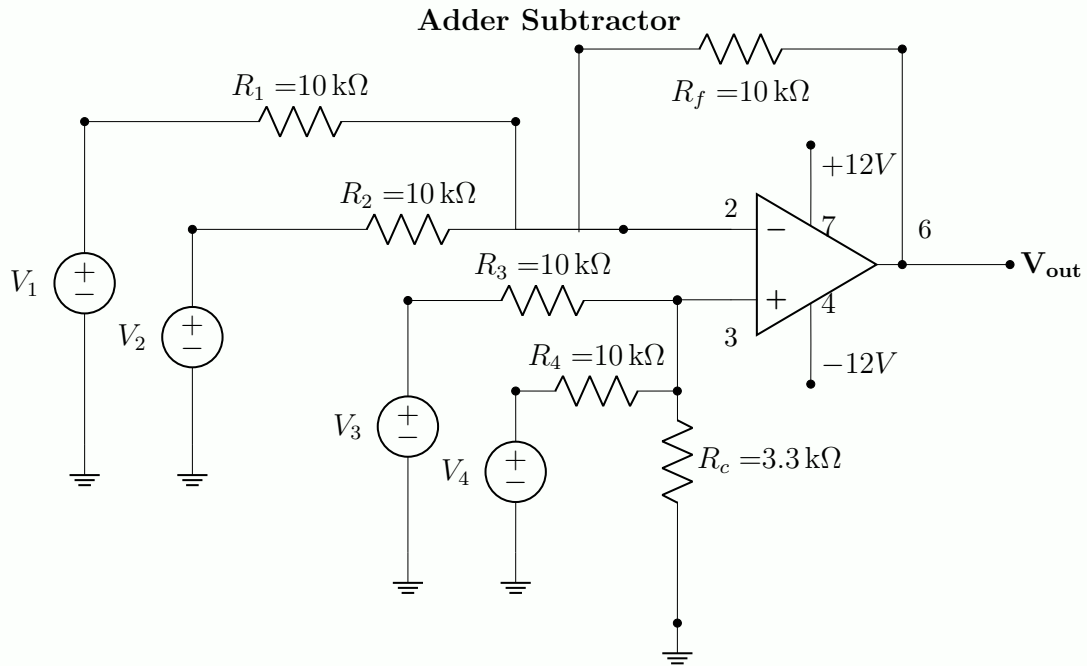
$$R_f = 3.3\,k\Omega$$

(rounded to the nearest standard resistor value)

Circuit Diagram



Circuit Diagram – Continued



Procedure

1. Connect the circuit as per the the configuration.
2. Apply the input voltages V_i to the input terminal of the operational amplifier.
3. Observe the output voltage and compare with the theoretical values .

Table 1: Op-Amp Summer

S.No	Inputs		Outputs			
	v_1 (V)	v_2 (V)	Theoretical Value (V)		Practical Value (V)	Output

Table 2: Op-Amp Subtractor

S.No	Inputs		Outputs	
	V_1 (V)	V_2 (V)	Theoretical (V)	Practical (V)

Table 3: Op-Amp Average Amplifier

S.No	Inputs			Outputs		
	V_1 (V)	V_2 (V)	V_3 (V)	Theoretical Value (V)	Practical Value (V)	Output

Table 4: Op-Amp Adder Subtractor

S.No	Inputs				Outputs	
	V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)	Theoretical Value (V)	Practical Output (V)

Pre lab Viva Questions

1. How would you choose resistor values in an inverting adder to ensure equal weighting for each input signal?
2. Given two input voltages, how would you design a subtractor using an op-amp to get an output proportional to $V_2 - V_1$?
3. How do you configure an op-amp circuit that can simultaneously perform addition of some inputs and subtraction of others?
4. What resistor arrangement is needed in an op-amp adder to produce an average (not sum) of multiple inputs?
5. Why must the op-amp be configured with feedback for these operations, and how does it influence linearity?

Post lab Viva Questions

1. If the output of a 3-input adder is lower than expected, how would you analyze and identify the fault in resistor selection?
2. In a subtractor circuit, if the output shows a large offset or incorrect polarity, what possible mistakes might have occurred in wiring or resistor ratio?
3. During testing of an adder-subtractor circuit, you observe the subtraction part working, but addition failing — how would you isolate and fix this?
4. How would you analyze whether your average amplifier is correctly computing the mean of four input voltages in the presence of resistor mismatch?
5. Analyze and compare the error sensitivity between the subtractor and average amplifier when there's a 5% tolerance in resistor values.

Result

Thus the adder, subtractor, adder-subtractor, and averaging amplifier circuits were designed and implemented using the IC 741 op-amp. Their outputs were verified, and their relevance to applications such as sensor interfacing, audio mixing, and instrumentation was understood

Ex No:3

DESIGN AND IMPLEMENTATION OF INTEGRATOR AND DIFFERENTIATOR CIRCUITS USING OP-AMP

Date:

Objective

To design and analyze op-amp-based integrator and differentiator circuits, and to understand their role in real-time analog signal processing tasks such as waveform generation, edge detection, signal filtering, and control systems..

Components /Equipments Required

Sl. No:	Components	Specification/Range	Qty.
1	Operational Amplifier	IC 741	1
2	Resistors	As per Design	As Required
3	Capacitors	As per Design	As Required
4	Regulated Power Supply	+ 12V, Dual	1
5	Function Generator	(1Hz – 3MHz.)	1
6	Cathode Ray Oscilloscope	30MHz, Dual Trace	1
7	Connecting Wires & CRO Probes	As Required	
8	Bread Board	-	1

Theory

Differentiator

A differentiator circuit provides an output voltage proportional to the rate of change (derivative) of the input voltage. In this configuration, the input voltage $V_{in}(t)$ is applied through a capacitor C to the inverting input of the op-amp, while a resistor R is connected in the feedback path from the output to the inverting input. The non-inverting terminal is grounded.

Using ideal op-amp assumptions (infinite gain and input impedance), the voltage at the inverting terminal is virtually zero (virtual ground). The current flowing through the capacitor is given by:

$$i_C = C \frac{dV_{in}(t)}{dt} \quad (1)$$

This current flows through the feedback resistor R , developing an output voltage V_{out} such that:

$$V_{out} = -Ri_C = -RC \frac{dV_{in}(t)}{dt} \quad (2)$$

Theory – Continued

Thus, the output is an inverted scaled derivative of the input. The circuit behaves like a high-pass filter, amplifying high-frequency components and attenuating low-frequency signals. This feature makes it useful in detecting fast transitions in pulse circuits, FM demodulation, and the derivative term of PID controllers.

Integrator Circuit

An integrator circuit produces an output voltage proportional to the time integral of the input signal. The input voltage is applied through a resistor R to the inverting terminal, and a capacitor C is placed in the feedback path. The non-inverting terminal is connected to ground.

Assuming ideal op-amp behavior and applying Kirchhoff's current law at the inverting node, the current through the resistor is:

$$i_R = \frac{V_{in}(t)}{R} \quad (3)$$

Since no current enters the op-amp input, this current must charge the capacitor, leading to:

$$i_C = C \frac{dV_{out}(t)}{dt} \quad (4)$$

Equating equations (3) and (4), we get:

$$\frac{V_{in}(t)}{R} = C \frac{dV_{out}(t)}{dt} \Rightarrow \frac{dV_{out}(t)}{dt} = \frac{1}{RC} V_{in}(t) \quad (5)$$

Integrating both sides with respect to time:

$$V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt \quad (6)$$

The negative sign indicates inversion. This circuit suppresses high-frequency noise and is widely used for ramp signal generation, smoothing, analog computing, and as the integral term in PID controllers.

Applications of Differentiator Circuit

- **Edge Detection in Pulse Circuits**

Detects rapid voltage transitions by generating spikes at rising/falling edges.

- **FM Demodulators**

Converts frequency variations in FM signals into amplitude changes.

- **Seismic and Vibration Monitoring**

Detects sudden changes from accelerometers or geophones (used in earthquake sens-

Theory – Continued

ing).

- **PID Controllers (Industrial Automation)**

Acts as the derivative term to predict and counteract error trends.

- **High-Pass Filtering**

Removes low-frequency noise from signals, useful in audio and instrumentation.

Applications of Integrator Circuit

- **Ramp Generator**

Produces a linearly increasing voltage from a step or square wave input.

- **Analog Computers**

Used in simulation and modeling to solve differential equations.

- **Analog Signal Conditioning**

Smooths noisy sensor outputs like temperature or pressure signals.

- **Analog-to-Digital Converters (ADC)**

Used in dual-slope ADCs for precise voltage-to-digital conversion.

- **Low-Pass Filter Applications**

Eliminates high-frequency noise in biomedical or instrumentation signals (e.g., ECG/EEG).

Design Calculation

Differentiator Circuit Using Op-Amp (IC 741)

Specifications: Desired lower cutoff frequency $f_a = 0.1 \text{ kHz}$ Operating signal frequency $f_b = 1 \text{ kHz}$ Capacitor $C_1 = 0.1 \mu F$

Step 1: Calculate Feedback Resistor R_F

The cutoff frequency f_a is determined by:

$$f_a = \frac{1}{2\pi R_F C_1}$$

$$R_F = \frac{1}{2\pi f_a C_1} = \frac{1}{2\pi \times 0.1 \times 10^3 \times 0.1 \times 10^{-6}}$$

$$R_F = \frac{1}{6.2832 \times 10^{-5}} \approx 15.923 \text{ k}\Omega \approx \boxed{15 \text{ k}\Omega}$$

Step 2: Calculate Input Resistor R_1

For operating frequency $f_b = 1 \text{ kHz}$:

$$f_b = \frac{1}{2\pi R_1 C_1}$$

$$R_1 = \frac{1}{2\pi \times 1 \times 10^3 \times 0.1 \times 10^{-6}} = \frac{1}{6.2832 \times 10^{-5}} \approx 1.592 \text{ k}\Omega \approx \boxed{1.5 \text{ k}\Omega}$$

Step 3: Verify Time Constant Consistency

Using the time constant equality:

$$R_1 C_1 = R_F C_F \Rightarrow C_F = \frac{R_1 C_1}{R_F}$$

$$C_F = \frac{1.5 \times 0.1}{15} = \boxed{0.01 \mu F}$$

Step 4: Equivalent Parallel Resistance R_{comp}

$$R_{\text{comp}} = R_1 \parallel R_F = \left(\frac{1}{R_1} + \frac{1}{R_F} \right)^{-1}$$

$$R_{\text{comp}} = \left(\frac{1}{15000} + \frac{1}{15000} \right)^{-1} = \boxed{1.5 \text{ k}\Omega}$$

Integrator Design

Lower frequency limit $f_a = 150 \text{ Hz}$

Choose capacitor $C_f = 0.01 \mu F$ (standard value)

Design – Continued**Formula:**

$$f_a = \frac{1}{2\pi R_f C_f} \Rightarrow R_f = \frac{1}{2\pi f_a C_f}$$

Calculation of Feedback Resistor R_f :

$$R_f = \frac{1}{2\pi \times 150 \times 0.01 \times 10^{-6}} = \frac{1}{9.42 \times 10^{-4}} \approx \boxed{106 \text{ k}\Omega}$$

Use the closest standard resistor:

$$\boxed{R_f = 100 \text{ k}\Omega}$$

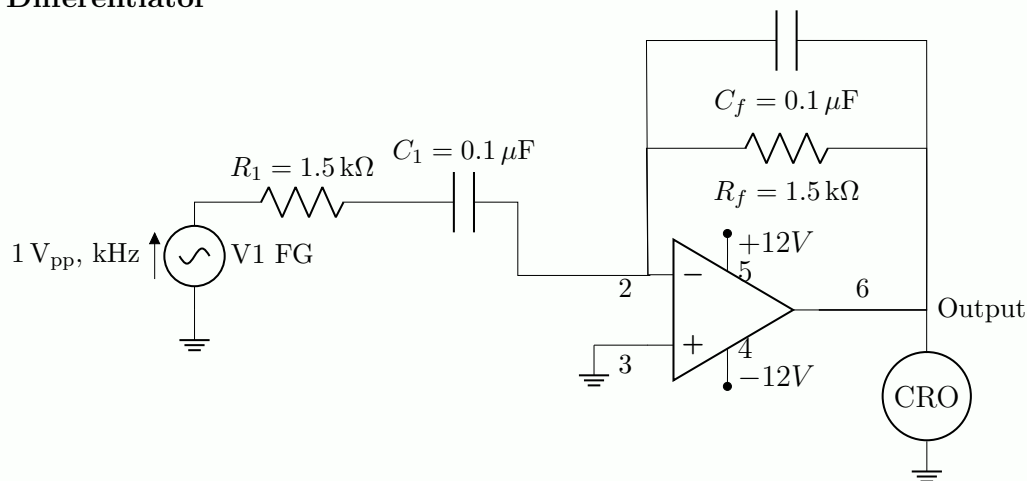
Selection of Input Resistor R_1 :

Design rule:

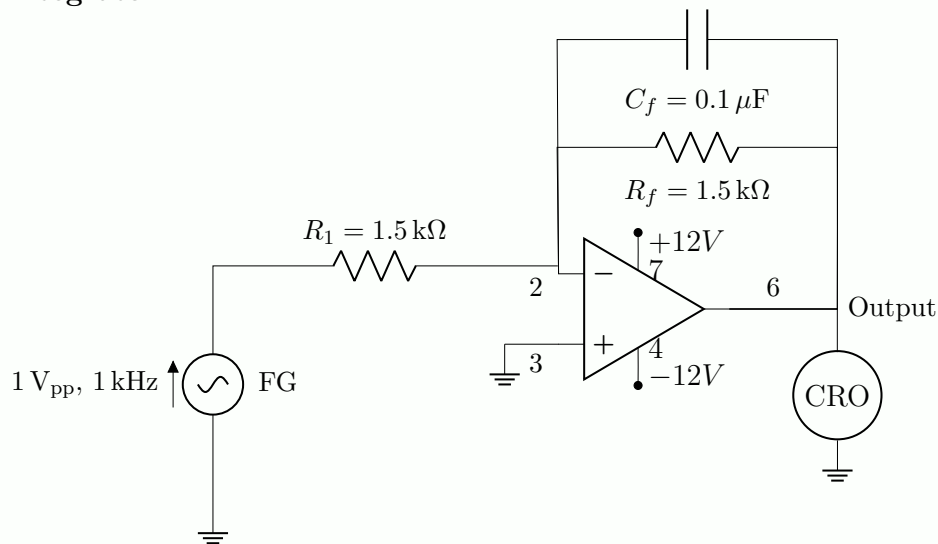
$$R_f = 10 \times R_1 \Rightarrow R_1 = \frac{R_f}{10} = \frac{100 \text{ k}\Omega}{10} = \boxed{10 \text{ k}\Omega}$$

Circuit Diagram

Differentiator



Integrator

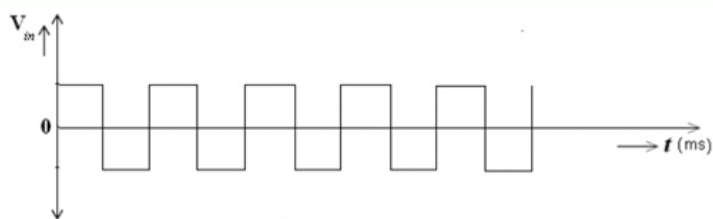


Procedure :

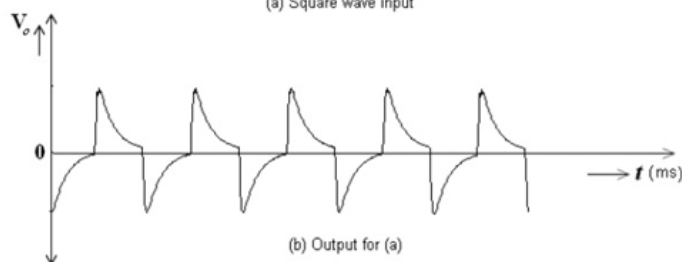
1. Connect the Integrator circuit on the breadboard as per the circuit diagram.
2. Power the IC 741 op-amp using a $\pm 12V$ dual power supply.
3. Apply a square wave input (1KHz 1V-pp) from a function generator.
4. Observe and record the output waveform using a CRO or DSO.
5. Repeat the above steps for the Differentiator circuit using the respective configuration.
6. For both circuits, apply sine and square waveforms as inputs and observe the corresponding outputs.

Model Graph

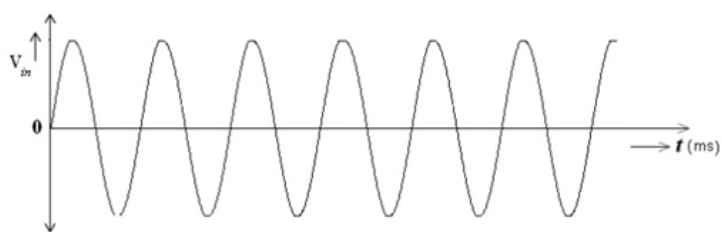
Differentiator



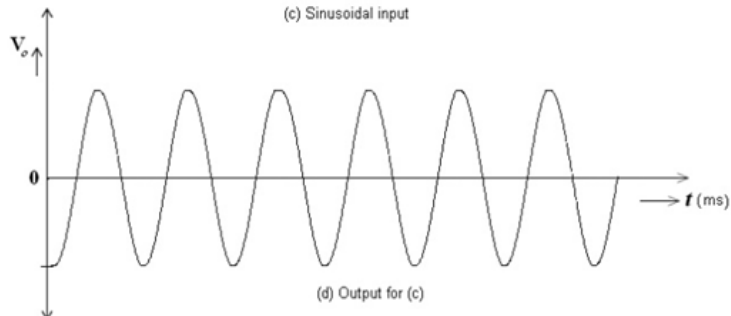
(a) Square wave input



(b) Output for (a)



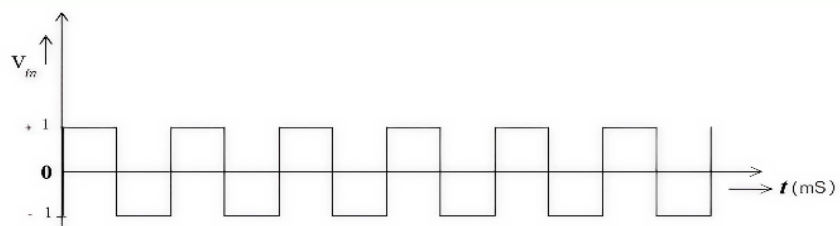
(c) Sinusoidal input



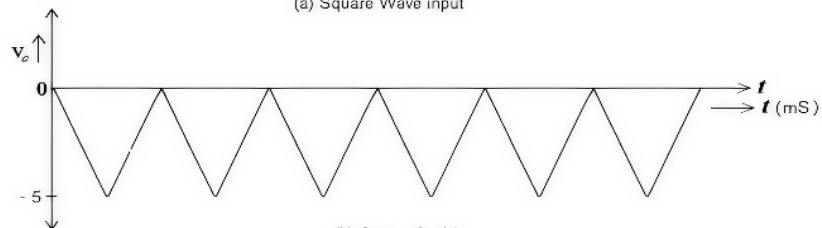
(d) Output for (c)

Model Graph– Continued

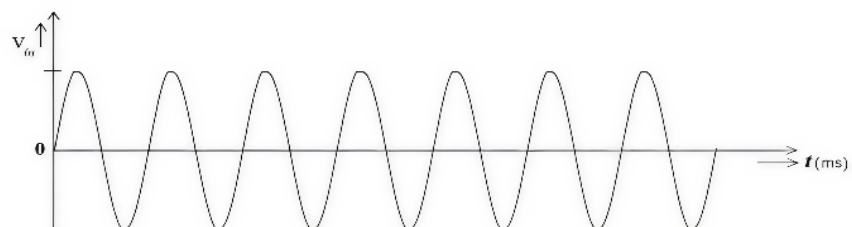
Integrator



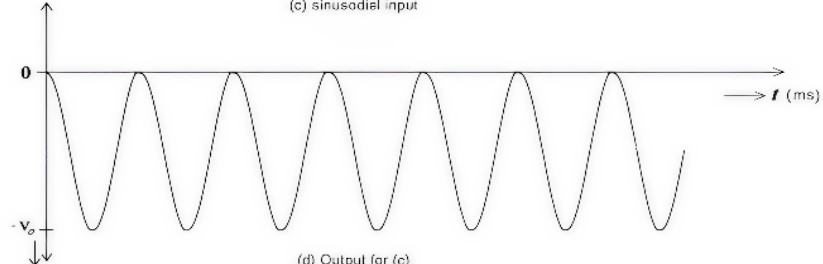
(a) Square Wave input



(b) Output for (a)



(c) sinusoidal input



(d) Output for (c)

Tabulation

Integrator					
S.No	Input Signal	V _{in} (V)	Frequency (Hz)	V _{out} (V)	Output Wave-form Description
1	Square Wave				
2	Sine Wave				

Tabulation – Continued

S.No	Input Signal	Vin (V)	Differentiator		Output Wave- form Description
			Frequency (Hz)	Vout (V)	
1	Square Wave				
2	Sine Wave				

Pre lab Viva Questions

Integrator

1. Apply the concept of Ohm's law to determine the voltage across the resistor in an op-amp integrator circuit.
2. How would you configure a function generator to apply a square wave input to an integrator circuit?
3. Given a resistor of $1.5\text{ k}\Omega$ and a capacitor of $0.1\text{ }\mu\text{F}$, calculate the time constant for the integrator.
4. Apply the concept of virtual ground to explain why current flows through the feedback path in an inverting integrator.
5. Sketch and interpret the expected output waveform when a square wave is applied to an ideal integrator.

Differentiator

1. Apply the capacitor charging equation to describe current flow in a differentiator input stage.
2. Calculate the expected output of a differentiator when a ramp signal is applied at the input.
3. How would you choose R and C values to limit noise amplification in a practical differentiator?
4. Explain the role of feedback resistance in stabilizing the differentiator's performance.
5. Sketch and label the output waveform of an op-amp differentiator when a square wave is applied at the input.

Post lab Viva Questions

Integrator (Op-Amp)

1. Based on your observations, how does the output waveform differ from the ideal integrator response and why?
2. Given a distorted output, how would you modify the component values to improve integration accuracy?
3. How would you apply the integrator in a triangular waveform generator circuit?
4. Explain how a low-frequency input affects the integrator output compared to a high-

Post lab Viva Questions – Continued

frequency input.

5. Use your experimental data to determine whether the integrator operated within its linear region.

Differentiator (Op-Amp)

1. Use your experimental results to explain the presence of sharp spikes in the differentiator output.
2. How would you apply a differentiator in an edge detection circuit?
3. Analyze how parasitic capacitance might affect the accuracy of your differentiator.
4. Based on your data, determine the time constant and validate it with the theoretical value.
5. Apply your understanding of frequency response to explain the differentiator's behavior at high frequencies.

Result

The Integrator and Differentiator circuits are implemented using IC 741.

- The Integrator produced ramp and smooth waveforms for square and sine inputs, respectively.
- The Differentiator generated sharp spikes for square inputs, indicating edge detection.

The observed outputs closely matches theoretical behaviour, confirming the circuits use in real-time applications like signal processing, waveform generation, and filtering.

Ex No: 4

Date:

DESIGN AND IMPLEMENTATION OF INSTRUMENTATION AMPLIFIER USING OP-AMP

Objective:

To design, construct, and verify the output voltage of an Instrumentation Amplifier using three IC 741 operational amplifiers, and to understand its applications in precise and differential signal amplification.

Components /Equipments Required

Sl. No	Components	Specification/Range	Qty.
1	Operational Amplifier	IC 741	3
2	Resistors	As per design	As required
3	Bread Board	-	1
4	Regulated Power Supply	+ 12V, Dual	1
5	Multimeter	-	1
7	Connecting Wires	As Required	

Theory

Theory and Design Model of Instrumentation Amplifier

Theory:

The instrumentation amplifier is a precision amplifier used for amplifying small differential signals while rejecting common-mode noise. It is widely used in biomedical signal processing and sensor-based applications. The amplifier consists of three op-amps: two in the first stage and one in the second.

Introduction: An **Instrumentation Amplifier (INA)** is a precision differential amplifier designed to amplify small differential signals while rejecting large common-mode voltages. INAs are preferred in measurement systems due to their high accuracy, high input impedance, low output impedance, and excellent Common-Mode Rejection Ratio (CMRR).

Architecture: The most common INA configuration uses three operational amplifiers:

- **Input Buffer Stage (2 Op-Amps):** This stage provides high input impedance, making it suitable for high-impedance sources such as sensors. It prevents loading effects and sets the gain using a single external resistor R_{gain} , which simplifies circuit design.
- **Difference Amplifier Stage (1 Op-Amp):** This stage subtracts the buffered signals and removes common-mode voltage. It is typically built using four equal

Theory – Continued

resistors, giving it a unity gain and high common-mode rejection.

Features:

- Gain is set by one external resistor R_{gain}
- High input impedance and low output impedance
- High common-mode rejection
- Suitable for amplifying low-level signals

Applications:

- Patient monitoring systems
- Battery testing equipment
- Analog input modules
- Motor control systems
- Industrial sensor interfacing

Design Calculation

Given Specifications: Desired Gain, $A_V = 5$

Choose: $R_1 = R_2 = 10 \text{ k}\Omega \Rightarrow \left(\frac{R_2}{R_1}\right) = 1$

Simplify the Gain Equation:

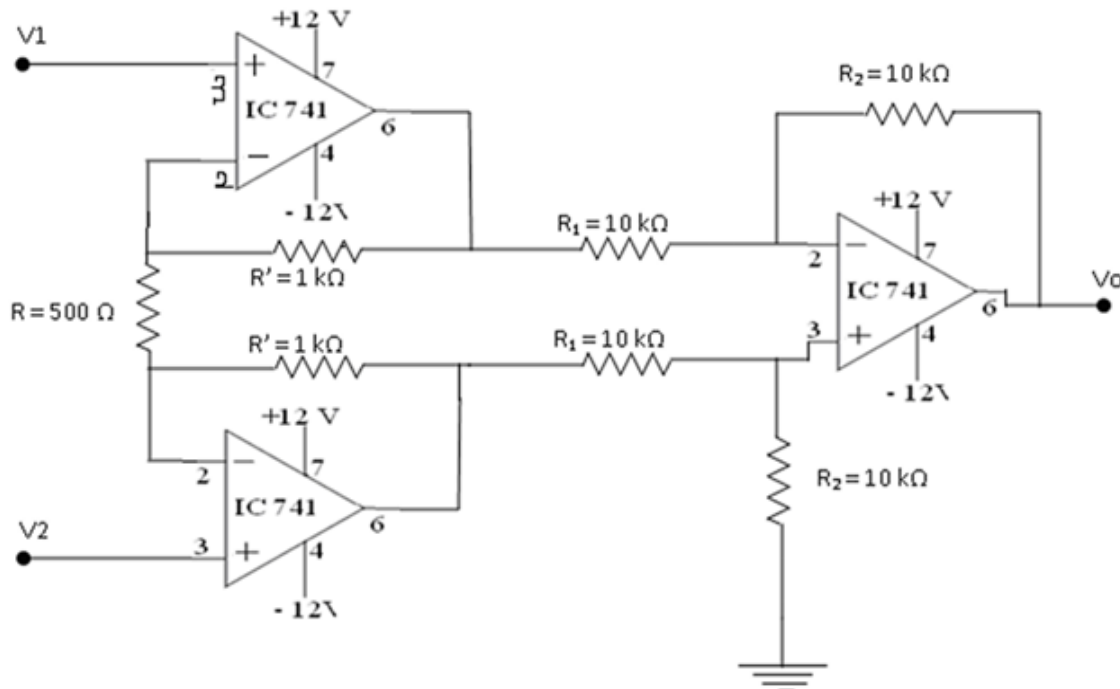
$$A_V = \left(\frac{R_2}{R_1}\right) \left(1 + \frac{2R'}{R}\right) = 1 \times \left(1 + \frac{2R'}{R}\right) \Rightarrow 5 = 1 + \frac{2R'}{R} \Rightarrow 4 = \frac{2R'}{R} \Rightarrow R = \frac{R'}{2}$$

Resistor Selection: Choose $R' = 1 \text{ k}\Omega \Rightarrow R = \frac{1 \text{ k}\Omega}{2} = 500 \Omega$

Gain Verification:

$$A_V = 1 + \frac{2 \times 1000 \Omega}{500 \Omega} = 1 + 4 = 5 \quad (\text{As required})$$

Circuit Diagram



Procedure

1. Connect the instrumentation amplifier circuit on a breadboard using three IC 741 op-amps and resistors $R = 500 \Omega$, $R' = 1 \text{ k}\Omega$, $R_1 = R_2 = 10 \text{ k}\Omega$.
2. Power the op-amps using a dual DC power supply of $\pm 12 \text{ V}$.
3. Apply input signals V_1 and V_2 from function generators to the amplifier inputs.
4. Measure the output voltage V_o using a voltmeter.
5. Vary the input difference $V_2 - V_1$ and observe the output to verify the gain and differential behavior.

S.No	R (k Ω)	Input VOLT- age V1 (volts)	Input VOLT- age V2 (volts)	Output Voltage (Vo)	Gain (Theoretical)	Gain (Practical)

Pre lab Viva Questions

1. How would you connect three op-amps to form an instrumentation amplifier?
2. Apply Ohm's law to determine the gain-setting resistor value in the amplifier circuit.
3. What configuration ensures high input impedance in an instrumentation amplifier?
4. How will you apply the concept of differential mode and common mode signals in this circuit?
5. How can you modify the gain of the instrumentation amplifier without disturbing its input stages?

Post lab Viva Questions

1. How did you verify the gain of the instrumentation amplifier practically?
2. If you replace the gain resistor R_G with a potentiometer, how does it affect performance?
3. How would you use an instrumentation amplifier to amplify a sensor output with small voltage levels?
4. If one op-amp fails in the input stage, how will the overall behavior of the amplifier change?
5. How can you apply this amplifier circuit in biomedical instrumentation systems like ECG or EMG?

Result

The instrumentation amplifier is designed and implemented using three IC 741 op-amps. The observed output voltages closely matched the theoretical values, validating the amplifier's ability to amplify differential signals with high precision and noise rejection.

Ex No:5	DESIGN AND IMPLEMENTATION OF WAVEFORM GENERATOR USING OP-AMP.
Date:	

Objective

To design and implement Astable multivibrator circuits using IC 741 operational amplifier, analyze their timing characteristics, and plot the corresponding waveforms. The experiment also demonstrates practical applications such as waveform generation, pulse shaping, and timing control in digital electronics and communication systems.

Components /Equipments Required

Sl. No	Components	Specification/Range	Qty.
1	Operational Amplifier	IC 741	3
2	Resistors	As per design	As Required
3	Capacitor	As per design	As Required
4	Bread Board	-	1
5	Regulated Power Supply	+ 12V, Dual	1
6	CRO	30 MHz	1
7	Connecting Wires	As Required	

Theory

Square Wave Generator

A square wave generator using an op-amp functions as a **relaxation oscillator**. The circuit comprises:

- A resistor and capacitor network connected to the inverting terminal.
- A voltage divider (feedback network) connected to the non-inverting terminal.

Working Principle:

- The capacitor charges and discharges alternately through the resistor.
- The op-amp output switches between positive and negative saturation voltages whenever the capacitor voltage crosses the threshold set by the voltage divider.
- This results in a continuous square waveform at the output.

The frequency of oscillation is given by:

$$f = \frac{1}{2RC \ln \left(\frac{1+\beta}{1-\beta} \right)}$$

where β is the fraction set by the voltage divider.

Theory – Continued

Applications of Square Wave Generator:

- Clock and timing circuits
- Switching waveform generators
- Pulse-width modulation (PWM) control
- Digital circuit testing
- Frequency measurement instruments

Triangular Wave Generator

The triangular wave generator consists of two op-amps:

- First op-amp (A1) acts as a **comparator** generating a square wave.
- Second op-amp (A2) acts as an **integrator** converting the square wave into a triangular wave.

Working Principle:

- The comparator switches between $+V_{\text{sat}}$ and $-V_{\text{sat}}$.
- This square wave is fed into the integrator, which integrates the voltage over time, producing a linearly increasing and decreasing ramp — i.e., a triangular wave.

Applications of Triangular Wave Generator:

- Function generators
- Audio and signal modulation
- Sawtooth/triangle ramp generation
- Testing linearity of analog circuits
- Signal processing in analog synthesizers

Design Calculation :

Design Calculation for Op-Amp Square Wave Oscillator

Step 1: Assume Desired Frequency of Oscillation

Let the required output frequency be:

$$f = 1 \text{ kHz}$$

Step 2: Choose Feedback Divider Ratio Assume equal resistor values for the voltage divider:

$$R_1 = R_2 = 10 \text{ k}\Omega \Rightarrow \beta = \frac{R_2}{R_1 + R_2} = 0.5$$

Step 3: Use Standard Timing Capacitor Select:

$$C = 0.1 \mu\text{F}$$

Design – Continued**Step 4: Use Frequency Equation of Square Wave Oscillator**

The frequency of oscillation for the op-amp relaxation oscillator is given by:

$$f = \frac{1}{2RC \ln \left(\frac{1+\beta}{1-\beta} \right)}$$

Substitute $\beta = 0.5 \Rightarrow \ln \left(\frac{1+0.5}{1-0.5} \right) = \ln(3) \approx 1.0986$ **Step 5: Rearranging for R**

$$R = \frac{1}{2Cf \ln 3} = \frac{1}{2 \times 0.1 \times 10^{-6} \times 1000 \times 1.0986}$$

$$R \approx 4.55 \text{ k}\Omega$$

Step 6: Choose Standard Resistor Value

Select the nearest standard resistor:

$$R = 4.7 \text{ k}\Omega$$

Design Calculation for Op-Amp Triangular Wave Oscillator**Step 1: Assume Desired Frequency of Oscillation**

Let the required frequency of output waveform be:

$$f = 1 \text{ kHz}$$

Step 2: Define Output Voltage Relationship

The peak-to-peak output voltage is given by:

$$V_{o(pp)} = 2 \cdot \frac{R_2}{R_1} \cdot V_{\text{sat}}$$

Assuming dual supply $\pm 12 \text{ V}$, the op-amp's saturation voltage:

$$V_{\text{sat}} \approx 10 \text{ V}$$

Let the desired peak-to-peak output be:

$$V_{o(pp)} = 5 \text{ V}$$

Assume:

$$R_2 = 1 \text{ k}\Omega$$

Design – Continued**Step 3: Calculate R_1**

$$R_1 = \frac{2 \cdot V_{\text{sat}} \cdot R_2}{V_{o(pp)}} = \frac{2 \cdot 10 \cdot 1 \text{ k}\Omega}{5} = 4 \text{ k}\Omega$$

Select nearest standard resistor:

$$R_1 = 4.7 \text{ k}\Omega$$

Step 4: Choose Capacitor Value

Assume:

$$C = 0.1 \mu\text{F}$$

Step 5: Use Frequency Formula to Find R

The frequency of oscillation is given by:

$$f = \frac{R_1}{4CR_2R} \Rightarrow R = \frac{R_1}{4fCR_2}$$

Substitute values:

$$R = \frac{4.7 \times 10^3}{4 \times 1000 \times 0.1 \times 10^{-6} \times 1 \times 10^3} = \frac{4700}{0.4} = 11.7 \text{ k}\Omega$$

Select nearest standard resistor:

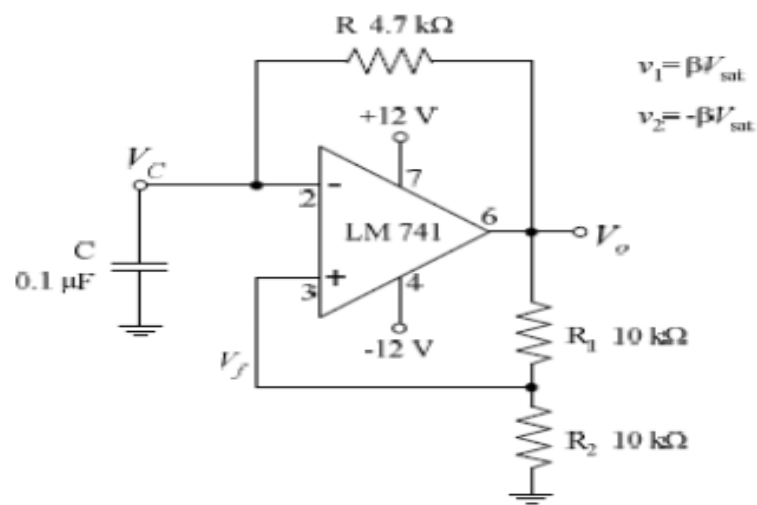
$$R = 12 \text{ k}\Omega$$

Procedure

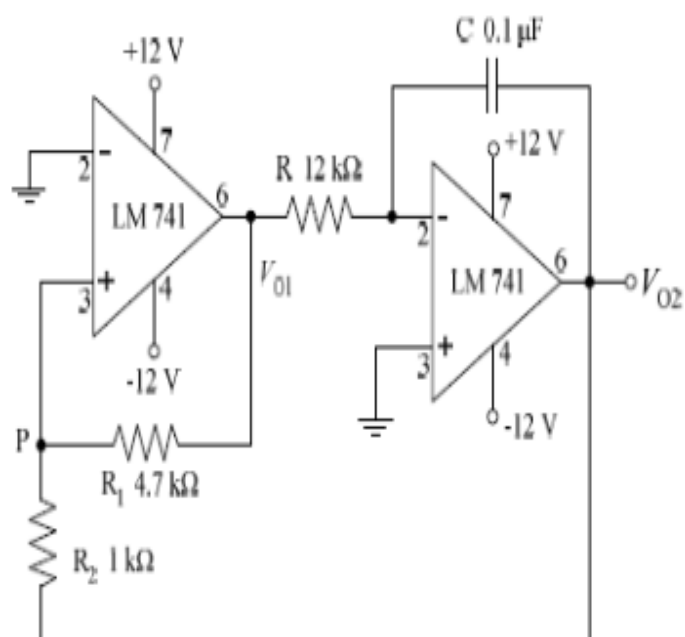
1. Set up the square wave generator circuit on the breadboard as shown in the circuit diagram. Observe the output waveform using a Cathode Ray Oscilloscope (CRO).
2. Record the amplitude and frequency of the square wave output observed on the CRO.
3. Set up the triangular wave generator circuit on the breadboard as shown in the corresponding circuit diagram. Observe the output waveform using the CRO.
4. Compare the observed frequencies of both the square and triangular waveforms with the theoretically designed frequency calculated during the design procedure.

Circuit Diagram

Square Wave Generator

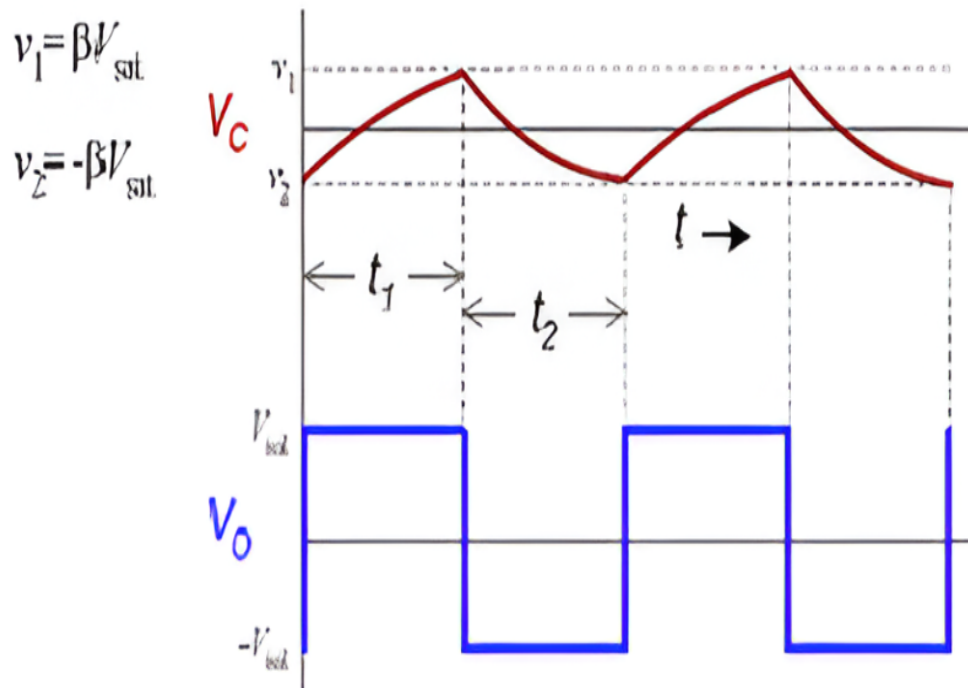


Triangular Wave Generator

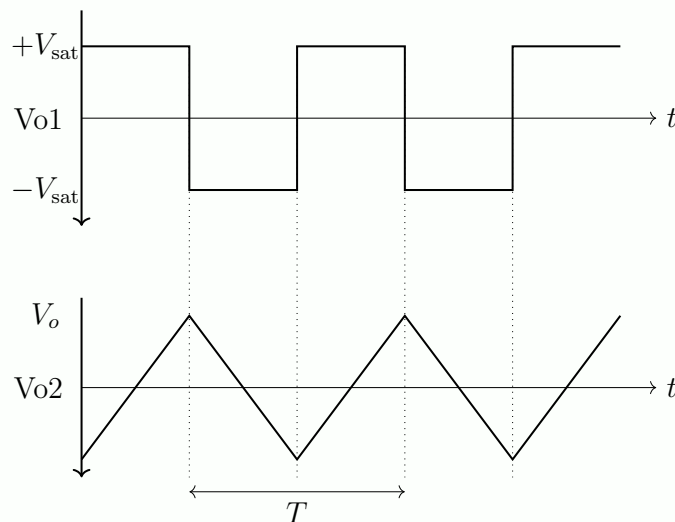


Model Graph

Output Waveforms: Capacitor voltage and V_o (Square Wave)



Output Waveforms: V_{o1} (Square Wave) and V_{o2} (Triangular Wave)



Pre lab Viva Questions

1. How would you configure an op-amp to function as a Schmitt trigger for generating square waves?
2. How can you generate a triangular wave using an op-amp integrator fed by a square wave?

Pre lab Viva Questions – Continued

3. What values of resistor and capacitor would you choose to get a frequency of 1 kHz in a square/triangle generator?
4. How would you adjust the duty cycle of the square wave in the generator circuit?
5. What will happen if you interchange the feedback resistor and the capacitor in the integrator stage?

Post lab Viva Questions:

1. How did you verify the frequency of the square and triangular waves in your circuit?
2. If the output triangular wave is distorted, how would you troubleshoot the integrator stage?
3. How does the feedback in the Schmitt trigger stage affect the output waveform?
4. How would you modify the circuit to vary the amplitude of the triangular wave without affecting frequency?
5. How can this waveform generator be applied in a practical analog function generator circuit?

Result

The square wave and triangular wave generator circuits are designed, constructed, and tested using op-amps. The output waveforms were observed and measured using a CRO. The results closely matched the designed frequency values. These waveform generators find applications in function generators, signal processing, waveform synthesis, and analog circuit testing..

Ex No:6	DESIGN AND ANALYSIS OF SECOND-ORDER ACTIVE FILTERS (LOW-PASS AND HIGH-PASS)
Date:	

Objective

To design, implement, and analyze second-order Butterworth active low-pass and high-pass filters using op-amp (IC 741) based on the Sallen-Key topology, and to study their frequency response characteristics in signal conditioning and analog filtering applications.

Components /Equipments Required

Sl. No	Components	Specification/Range	Qty.
1	Operational Amplifier	IC 741	3
2	Resistors	As per Design	As Required
3	Capacitor	As per Design	As Required
4	Bread Board	-	1
5	Regulated Power Supply	+ 12V, Dual	1
6	CRO	30 MHz	1
7	Connecting Wires	As Required	-

Theory

Filters are essential components in analog and digital signal processing systems. They are used to selectively allow signals of certain frequencies to pass while attenuating unwanted frequency components. Based on their frequency response, filters are broadly categorized into:

- Low-pass filters (LPF)
- High-pass filters (HPF)
- Band-pass filters (BPF)
- Band-stop filters (BSF)

Filters can be implemented using passive components (resistors, capacitors, and inductors) or active components such as operational amplifiers (op-amps). Active filters provide several advantages:

- Amplification capability
- Smaller size (no inductors)
- Better control over cutoff frequency
- Impedance matching

Theory – Continued

Butterworth Filters

The Butterworth filter is widely used due to its maximally flat magnitude response in the passband. This means the gain remains constant up to the cutoff frequency with no ripples, making it ideal for applications requiring undistorted signal transmission within the passband.

For a second-order Butterworth filter:

- The standard damping factor $\alpha = 1.414$, and the gain is

$$A_o = 3 - \alpha = 1.586$$

- The gain in the passband can be set using the feedback resistor network in an op-amp configuration.

$$A_o = 1 + \frac{R_f}{R_i}$$

Sallen-Key Topology

The Sallen-Key topology is a popular configuration for implementing second-order filters using an op-amp. It is known for:

- Simplicity
- Ease of tuning
- Ability to implement both low-pass and high-pass filters by reconfiguring the RC network

The cutoff frequency f_c for both LPF and HPF is determined by:

$$f_c = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

Low-Pass Filter (LPF)

A second-order LPF allows signals with frequencies below the cutoff to pass through while attenuating higher frequencies. It is used to:

- Remove high-frequency noise
- Prevent aliasing effects in analog systems

High-Pass Filter (HPF)

A second-order HPF allows signals above the cutoff frequency to pass while attenuating lower frequencies. Applications include:

- Removing DC offsets
- Filtering low-frequency drift from sensor signals

Theory – Continued

Key Features of Second-Order Butterworth Active Filters

- **Flat Passband Response:** No ripples; smooth magnitude response up to the cutoff frequency.
- **Sharp Roll-Off:** Attenuation rate of -40 dB/decade for second-order filters.
- **Tunable Cutoff Frequency:** Controlled using RC values.
- **Gain Control:** Adjustable using op-amp feedback network.
- **Compact Implementation:** Requires no inductors; suitable for integration.
- **Stable and Predictable Behavior:** Ideal for signal conditioning and analog filtering.

Applications

- Audio signal processing and equalization
- Biomedical instrumentation (ECG, EEG filtering)
- Anti-aliasing filters in analog-to-digital conversion
- Communication systems for channel separation
- Vibration and motion signal filtering
- Embedded systems for sensor signal conditioning

Design Calculation

Design Specifications -LPF

- Filter Type: Low-Pass Filter (LPF)
- Order: Second-order
- Response: Butterworth (maximally flat)
- Cutoff Frequency f_c : 1 kHz
- Topology: Sallen-Key
- Damping factor α : 1.414
- Passband Gain A_o : $3 - \alpha = 1.586$

Design Equations

For a Sallen-Key LPF, the cutoff frequency is:

$$f_c = \frac{1}{2\pi RC}$$

Gain:

$$A_o = 1 + \frac{R_f}{R_i} = 1.586 \Rightarrow \frac{R_f}{R_i} = 0.586$$

Design – Continued

Design Procedure

1. Choose Capacitor Values

Let,

$$C_1 = C_2 = 0.01 \mu\text{F}$$

2. Calculate Resistors

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \cdot 1000 \cdot 0.01 \times 10^{-6}} \approx 15.9 \text{ k}\Omega$$

Choose:

$$R_1 = R_2 = 15.9 \text{ k}\Omega \quad (\text{use } 16 \text{ k}\Omega \text{ standard value})$$

3. Set Op-amp Gain

$$R_i = \frac{R_f}{0.586}, \quad R_f = 10 \text{ k}\Omega \Rightarrow R_i = \frac{10 \text{ k}\Omega}{0.586} \approx 5.86 \text{ k}\Omega$$

Use a 5.6 k Ω or 6.2 k Ω resistor or a 10 k Ω potentiometer to fine-tune.

Design Specifications -HPF

$$f_c = \frac{1}{2\pi RC}$$

For Butterworth response, the gain A_o should be:

$$A_o = 3 - \alpha = 1.586$$

Set gain using:

$$A_o = 1 + \frac{R_f}{R_i} \Rightarrow \frac{R_f}{R_i} = 0.586$$

1. Choose Capacitor Values

Let,

$$C_1 = C_2 = 0.1 \mu\text{F}$$

2. Calculate Resistor Values

Using:

$$R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \cdot 1000 \cdot 0.1 \times 10^{-6}} \approx 1591.5 \Omega$$

Design – Continued

Round to the nearest standard value:

$$R_1 = R_2 = 1.6 \text{ k}\Omega$$

3. Set Op-Amp Gain Resistors

Choose:

$$R_i = 10 \text{ k}\Omega$$

- Then, $R_f = 0.586 \times R_i = 5.86 \text{ k}\Omega \approx 5.6 \text{ k}\Omega$ (standard E12)

You can use a **5.6 k Ω resistor** or **use a 10 k Ω potentiometer** to fine-tune.

Procedure

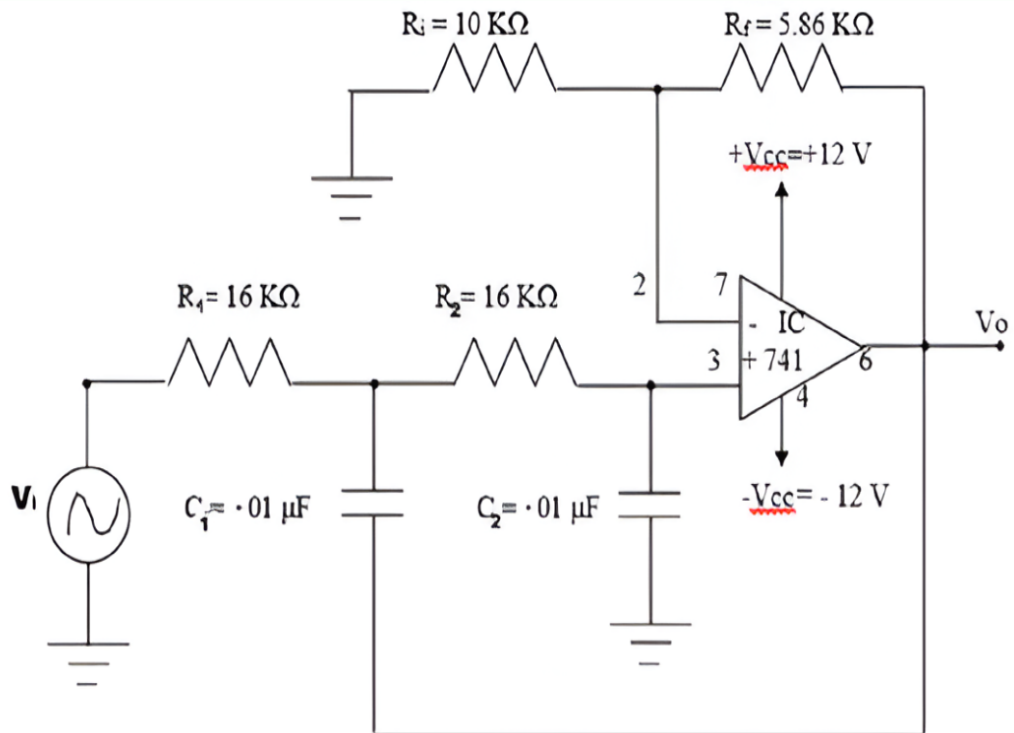
1. Connect the second-order low-pass filter circuit as per the given design on the bread-board.
2. Apply an input sine wave using a function generator.
3. Vary the frequency from low to high and measure the output voltage using a CRO or DSO.
4. Record the output voltage at different frequencies and calculate the voltage gain using:

$$A_o = \frac{V_o}{V_i}, \quad G_o = 20 \log_{10}(A_o)$$

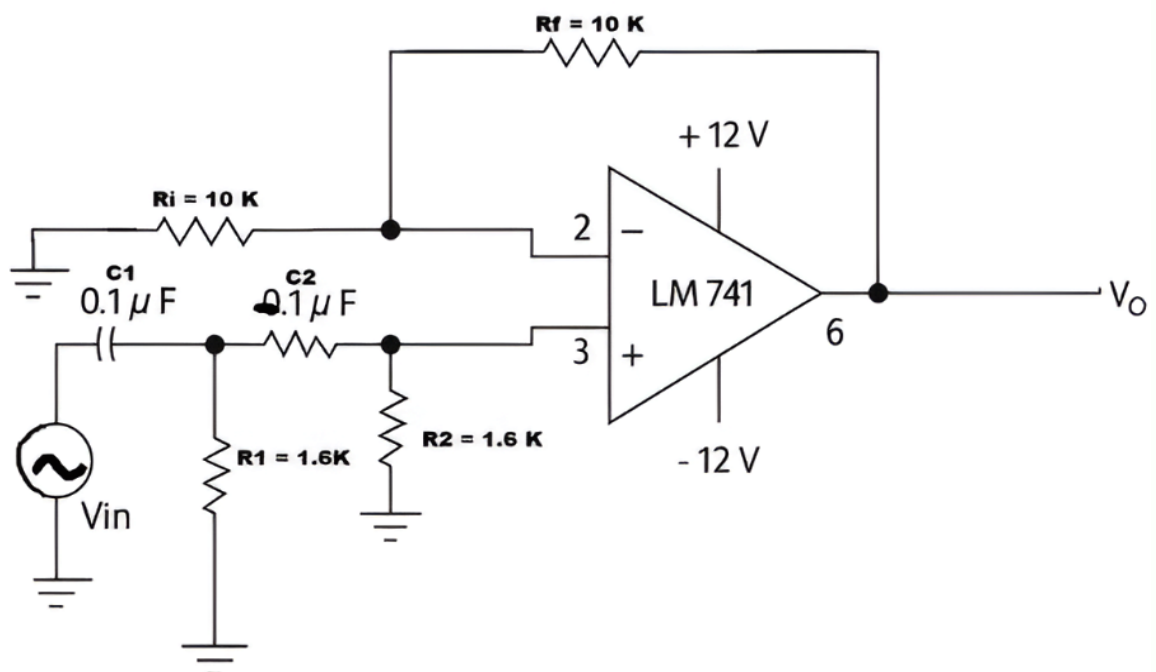
5. Repeat steps 1 to 4 for the second-order high-pass filter.
6. Plot the frequency response graphs (Gain in dB vs. Frequency) for both filters on a semi-log graph sheet.

Circuit Diagram

BUTTERWORTH SECOND ORDER ACTIVE LPF



BUTTERWORTH SECOND ORDER ACTIVE HPF



Tabulation

BUTTERWORTH SECOND ORDER ACTIVE LPF

Input Voltage : —V

S.No	Frequency (Hz)	Output Voltage (V) V_o	Voltage Gain $A_o = \frac{V_o}{V_i}$	Gain (dB)	Voltage Gain G_o

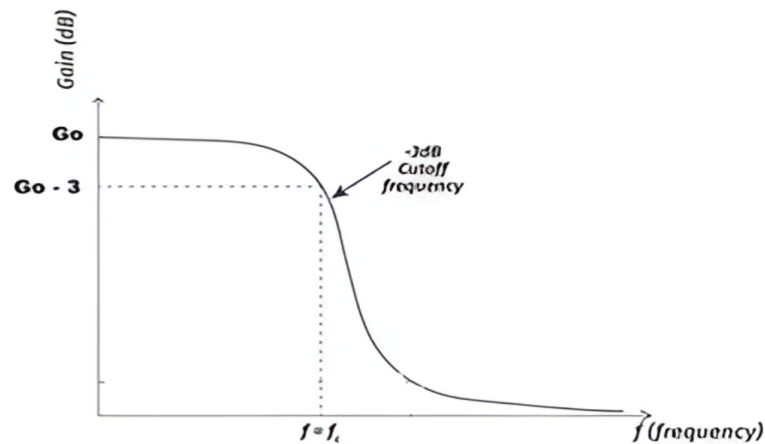
BUTTERWORTH SECOND ORDER ACTIVE HPF

Input Voltage : —V

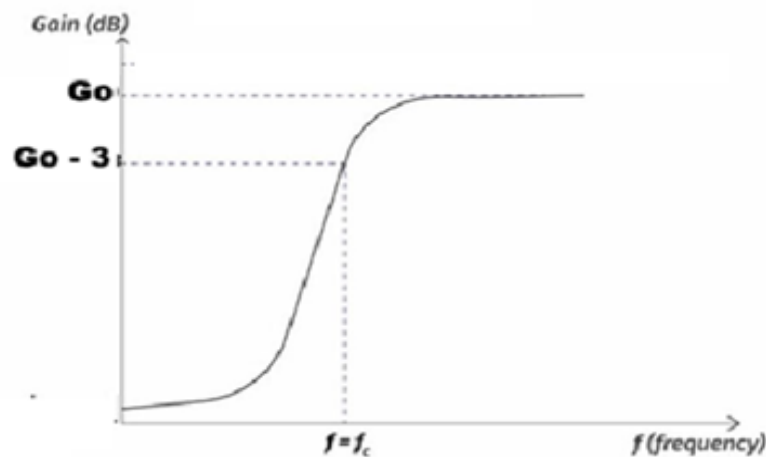
S.No	Frequency (Hz)	Output Voltage (V) V_o	Voltage Gain $A_o = \frac{V_o}{V_i}$	Gain (dB)	Voltage Gain G_o

Model Graph

BUTTERWORTH SECOND ORDER ACTIVE LPF



BUTTERWORTH SECOND ORDER ACTIVE HPF



Pre lab Viva Questions

1. How would you design a low-pass filter using an op-amp to achieve a specific cutoff frequency?
2. How can a high-pass filter be implemented using an op-amp and reactive components?
3. What values of R and C would you select to set a 1 kHz cutoff in a low-pass filter?
4. How would you modify a first-order filter to make it a second-order active filter?
5. How do you apply the concept of frequency response while analyzing filter circuits?

Post lab Viva Questions

1. How did you measure and verify the cutoff frequency of the filter practically?
2. How would you use a signal generator and CRO to analyze the filter's frequency

Post lab Viva Questions – Continued

response?

3. How did the observed output waveform change when the input frequency was above or below the cutoff frequency?
4. How can these filters be cascaded for band-pass or band-stop applications?
5. How would you use this filter circuit in audio or communication systems practically?

Result

The second-order Low-Pass and High-Pass active filters were designed and implemented using IC 741. The frequency response of the designed filters was observed, and the practical cutoff frequencies closely matched the theoretical calculations, thereby validating the expected filtering behavior and confirming the accuracy of the design.

Ex No:7	SIMULATION OF ANALOG TO DIGITAL CONVERTER(ADC)
Date:	

Objective

To simulate and analyze the operation of a 3-bit Flash Analog-to-Digital Converter (ADC) using comparators and resistor ladder in a simulation environment. The experiment demonstrates how analog input voltage is converted into a digital output using a thermometer code and basic comparator logic.

Software Required

NI Multisim Software 14.3

Theory

Analog-to-Digital Converters (ADCs) play a vital role in interfacing real-world analog signals with digital systems. They convert continuous-time, continuous-amplitude analog signals into discrete-time, discrete-amplitude digital representations that can be processed by microcontrollers, digital signal processors, and other digital devices.

Among various ADC architectures, the **Flash ADC** (also known as a **parallel ADC**) is the fastest and most suitable for high-speed applications such as digital oscilloscopes, radar systems, high-frequency data acquisition, and video processing.

Working Principle

A Flash ADC uses a set of comparators to compare the analog input voltage (V_{in}) with a series of reference voltages derived from a resistor ladder connected across a stable voltage source. Each comparator gives a digital HIGH (logic 1) if V_{in} is greater than its corresponding reference voltage and LOW (logic 0) otherwise.

This results in a **thermometer code**: a sequence of 1s followed by 0s, that indicates how many reference thresholds the input has crossed. This thermometer code is typically passed through a digital priority encoder to convert it into a binary output corresponding to the input level.

Implementation in Simulation

In this experiment, a **3-bit Flash ADC** is simulated using operational amplifiers configured as comparators in a **NI Multisim** simulation environment. A resistor ladder provides equally spaced reference voltages, and the comparators generate outputs that are visualized using LEDs or digital indicators. The number of active LEDs indicates the approximate range of the analog input.

Theory – Continued

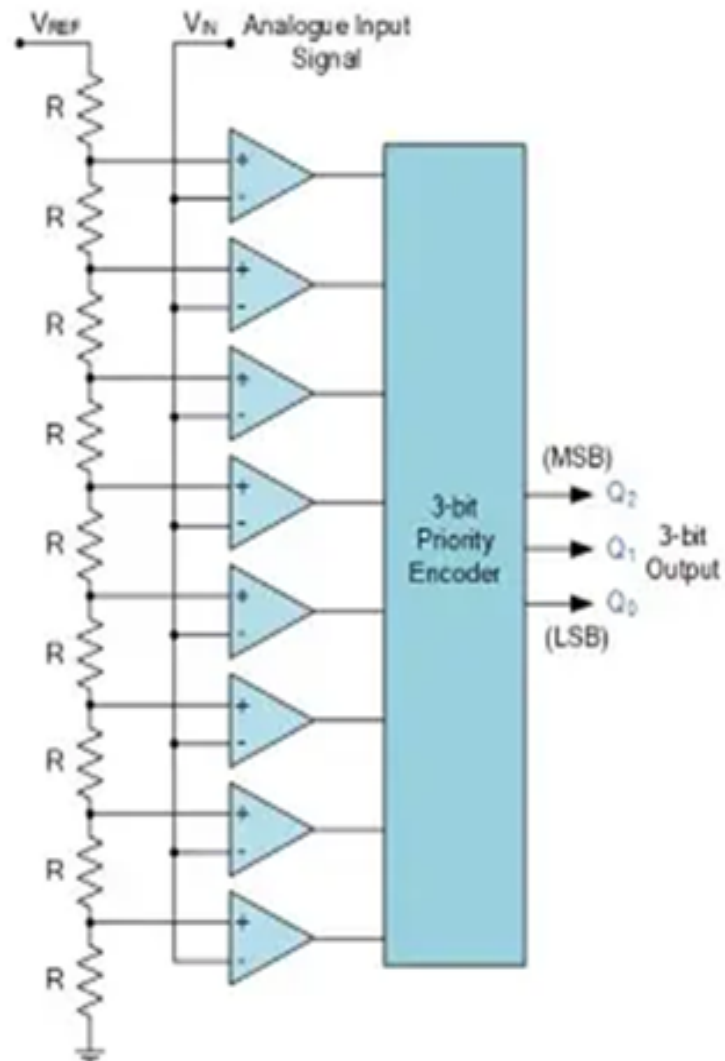
Key Features

- Ultra-fast conversion speed due to parallel comparator architecture
- Simple implementation using basic components like op-amps and resistors
- Visual output representation using LEDs to display comparator states
- Demonstrates key ADC concepts like reference voltage division, comparator triggering, thermometer code, and binary encoding

Applications

- High-speed digital oscilloscopes
- Video digitization and image processing systems
- Radar and communication receivers
- Real-time data acquisition in embedded systems
- Medical electronics (e.g., ECG, EEG digitization)
- Software-defined radios (SDRs)

Circuit Diagram



Design Calculation

To convert an analog voltage in the 0-5V range into a 3 bit digital output using a comparator-based Flash ADC architecture.

Resolution & Step Size Calculation

- Number of output bits: $n = 3$
- Number of levels: $2^n = 8$
- Number of comparators: $2^n - 1 = 7$
- Input range: 0 to 5 V

Design – Continued

- Step size:

$$\text{Step size} = \frac{\text{Input range}}{\text{Number of levels}} = \frac{5 \text{ V}}{8} = 0.625 \text{ V}$$

Component Design

1. Resistor Ladder Network

- Use 8 equal resistors ($1 \text{ K}\Omega$) in series between $V_{\text{ref}} = 5 \text{ V}$ and GND.
- Tap 7 intermediate nodes to generate reference voltages for comparators.
- Reference voltages:
 - $V_{R1} = 0.625 \text{ V}$
 - $V_{R2} = 1.25 \text{ V}$
 - $V_{R3} = 1.875 \text{ V}$
 - $V_{R4} = 2.5 \text{ V}$
 - $V_{R5} = 3.125 \text{ V}$
 - $V_{R6} = 3.75 \text{ V}$
 - $V_{R7} = 4.375 \text{ V}$

2. Comparators (Op-Amps)

- Non-inverting input: Analog input V_{in}
- Inverting input: Connected to each reference voltage from the resistor ladder
- Output:

HIGH (Logic 1) if $V_{\text{in}} > V_{\text{ref}}$, else LOW (Logic 0)

3. Priority Encoder

- Inputs: 7-bit thermometer code from comparators
- Output: 3-bit binary code corresponding to the number of comparators triggered

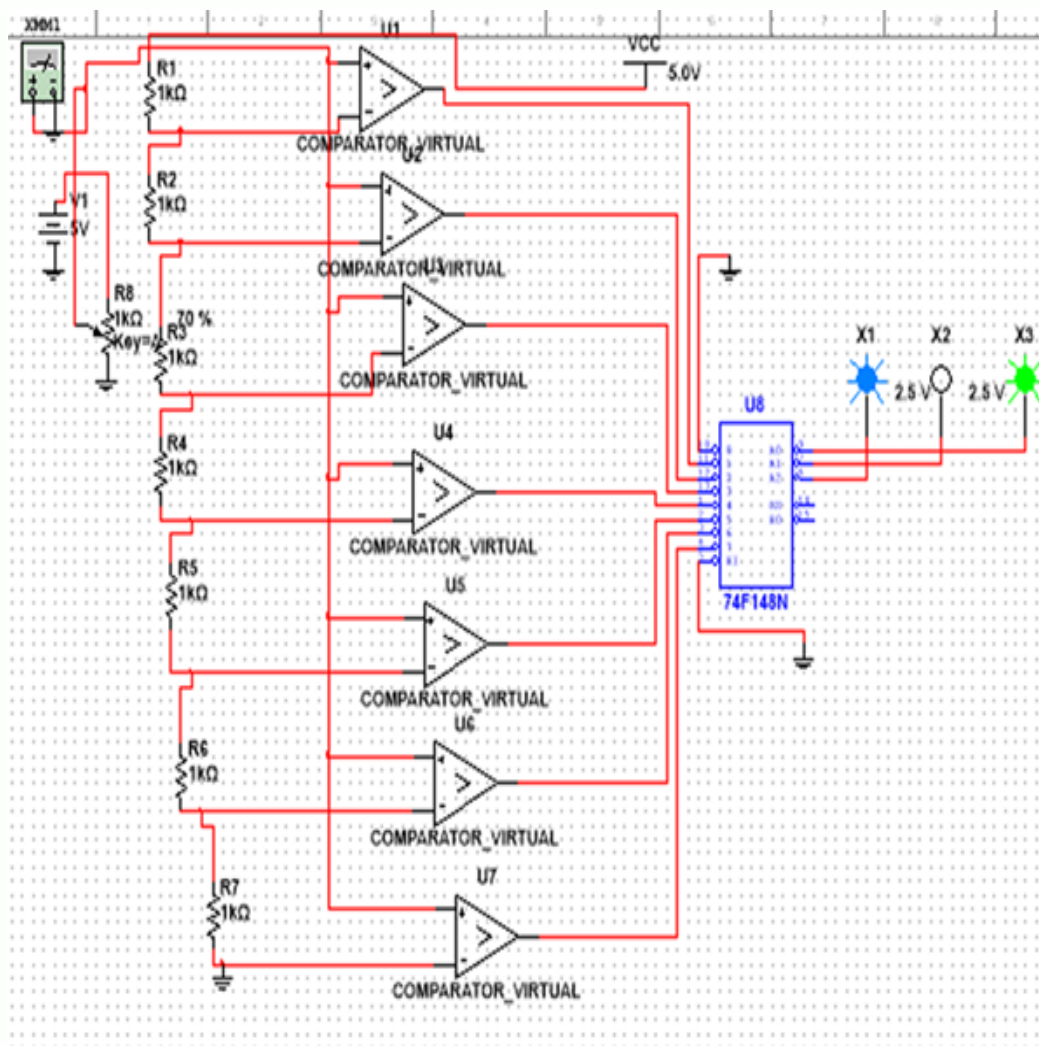
3-Bit Flash ADC Output Table

Analog Input V_{in} Range	Comparator Output (LEDs)	Thermometer Code	Binary Output
0 – 0.625 V	All OFF	0000000	000
0.625 – 1.25 V	0000001	0000001	001
1.25 – 1.875 V	0000011	0000011	010
1.875 – 2.5 V	0000111	0000111	011
2.5 – 3.125 V	0001111	0001111	100
3.125 – 3.75 V	0011111	0011111	101
3.75 – 4.375 V	0111111	0111111	110
> 4.375 V	1111111	1111111	111

Procedure

1. Launch **NI Multisim** (or any circuit simulation software) and open a blank schematic.
2. Create a resistor ladder using eight equal-value resistors ($1\text{ k}\Omega$) connected in series between a 5 V DC supply and ground.
3. Tap seven intermediate nodes from the resistor ladder to serve as reference voltages.
4. Place seven op-amps or comparator components (e.g., **LM741**) and configure each as a comparator.
5. Connect the inverting input of each comparator to one of the reference voltage taps from the resistor ladder.
6. Connect the non-inverting input of all comparators to a common analog input voltage source V_{in} , preferably a DC voltage source or voltage slider for easy variation.
7. Connect the output of each comparator to an LED or digital indicator to observe logic HIGH (1) or LOW (0) states.
8. Run the simulation.
9. Vary the analog input voltage from 0 V to 5 V in small steps (0.5 V) and observe how many comparators turn ON (i.e., output logic 1), forming a **thermometer code**.
10. Record the corresponding comparator outputs and binary values in the tabulation table.

Circuit Schematic



Tabulation

Observation Table – 3-Bit Flash ADC

S.No	V_{in} (V)	Comparator Output	Thermometer Code	Binary Output
1				
2				
3				
4				
5				
6				
7				

Pre lab Viva Questions

1. How would you represent an analog signal in digital form using an ADC?
2. How can you simulate an ADC circuit in Multisim? What components are typically required?
3. How do resolution and reference voltage affect ADC accuracy?
4. How would you choose the number of bits in an ADC for a specific application?
5. What changes would you make in the circuit to increase the sampling rate in the simulation?

Post lab Viva Questions

1. How did you verify the correctness of the ADC output in the Multisim simulation?
2. How did varying the analog input affect the digital output during simulation?
3. How would you troubleshoot if the ADC output in Multisim is not changing as expected?
4. How can this ADC circuit be interfaced with digital components in practical systems?
5. How can the observed simulation results help in designing real-time data acquisition systems?

Result

The 3-bit Flash ADC was successfully implemented and simulated. The comparator outputs matched the expected thermometer code, and the corresponding binary outputs were verified for varying analog input levels.

Ex No:8	SIMULATION OF DIGITAL TO ANALOG CONVERTER (DAC)
Date:	

Objective

To simulate and analyze the operation of a 4-bit Digital-to-Analog Converter (DAC) using a binary-weighted resistor network in NI Multisim. This experiment demonstrates the conversion of 4-bit digital input into a corresponding analog voltage.

Software Required

NI Multisim Software 14.3

Theory

Digital-to-Analog Converters (DACs) convert discrete digital signals into continuous analog voltages. They play a critical role in interfacing digital circuits with real-world analog devices such as speakers, sensors, and control actuators. The Binary-Weighted Resistor DAC is one of the simplest DAC architectures.

Working Principle

In a binary-weighted resistor DAC, each digital input bit controls a switch that applies either a reference voltage (logic HIGH = 5 V) or ground (logic LOW = 0 V) through a resistor whose value is inversely proportional to the bit's significance. These resistors connect to the inverting input of an op-amp configured as a summing amplifier. The op-amp sums the weighted currents and converts them into a proportional analog voltage.

The output voltage is given by:

$$V_{\text{out}} = -V_{\text{ref}} \times \left(\frac{D3}{16} + \frac{D2}{8} + \frac{D1}{4} + \frac{D0}{2} \right)$$

where $D3, D2, D1, D0 \in \{0, 1\}$ are the 4-bit digital inputs.

Implementation in Simulation

- Four digital switches are used to represent 4-bit digital input (D3 to D0).
- Resistors with binary-weighted values (8 kΩ, 4 kΩ, 2 kΩ, and 1 kΩ) connect each switch to the inverting terminal of an LM741 op-amp.
- A 1 kΩ feedback resistor is used to complete the summing amplifier configuration.
- The op-amp is powered with a dual ±12 V supply.
- The analog output is monitored using an oscilloscope or voltage probe in Multisim.

Theory – Continued

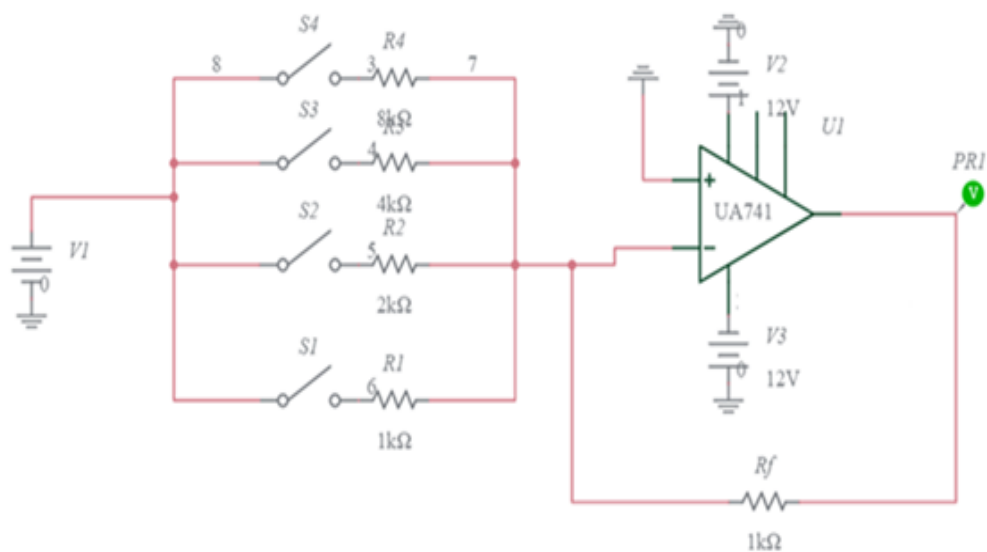
Key Features

- Simple DAC architecture using resistors and op-amp
- Converts 4-bit digital data to continuous analog voltage
- Easy to simulate and visualize digital-to-analog conversion
- Helps understand the concept of binary-weighted resistor networks

Applications

- Audio signal generation
- Analog control in embedded systems
- Digital waveform generation
- Signal reconstruction in communication systems
- Function generators and instrumentation systems

Circuit Diagram



Design Calculation

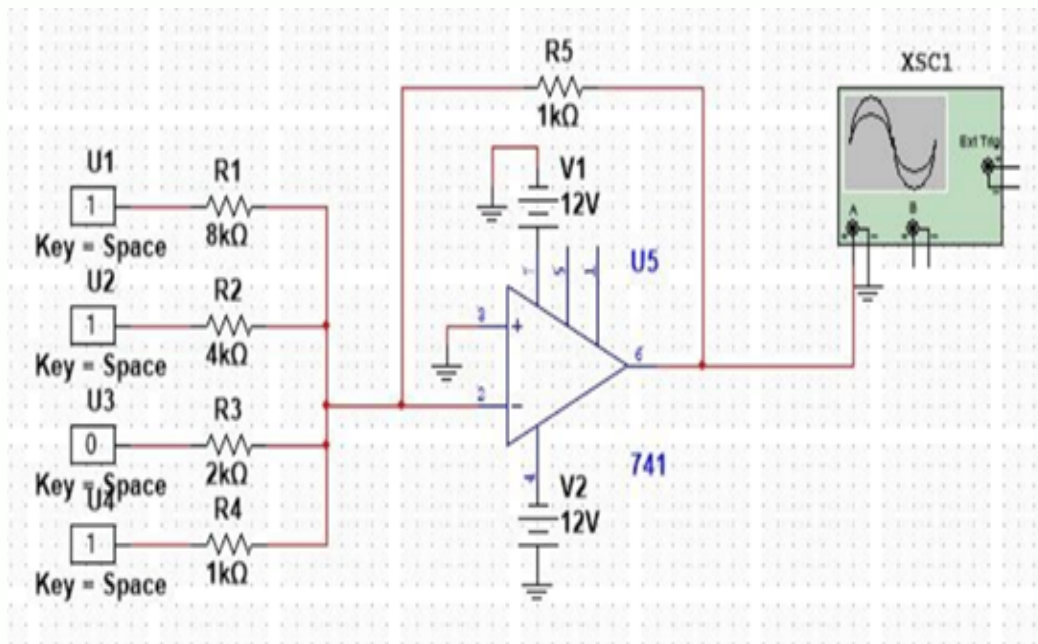
- $V_{\text{ref}} = 5\text{ V}$
- Feedback resistor $R_f = 1\text{ k}\Omega$
- Each bit's contribution:
 - D3 (MSB): $5\text{ V} \times \frac{1}{16} = 0.3125\text{ V}$
 - D2: $5\text{ V} \times \frac{1}{8} = 0.625\text{ V}$
 - D1: $5\text{ V} \times \frac{1}{4} = 1.25\text{ V}$
 - D0 (LSB): $5\text{ V} \times \frac{1}{2} = 2.5\text{ V}$

Design – Continued

- Maximum Output Voltage:

$$V_{\text{out max}} = -(0.3125 + 0.625 + 1.25 + 2.5) = -4.6875 \text{ V}$$

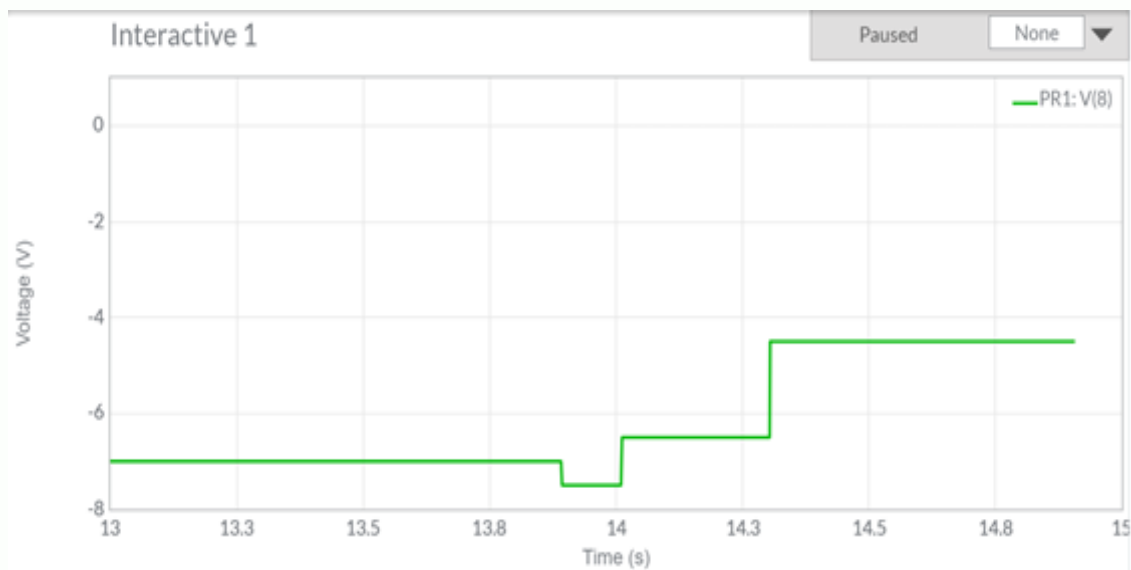
Circuit Schematic



Procedure

1. Open **NI Multisim** and start a new schematic.
2. Place four digital switches (D3 to D0) and label them accordingly.
3. Connect the switches through weighted resistors:
 - D3: 8 kΩ
 - D2: 4 kΩ
 - D1: 2 kΩ
 - D0: 1 kΩ
4. Connect the resistor ends to the inverting input of an LM741 op-amp.
5. Place a 1 kΩ feedback resistor from the op-amp output to its inverting input.
6. Ground the non-inverting input of the op-amp.
7. Power the op-amp with a dual supply of ±12 V.
8. Connect the op-amp output to an oscilloscope or voltage probe.
9. Simulate by toggling the digital inputs from 0000 to 1111 and record the corresponding output voltages.

Model Graph



Pre lab Viva Questions

1. How did the DAC output change when you varied the digital input during the simulation?
2. How did you confirm the linearity and resolution of the DAC output waveform?
3. How would you modify the DAC circuit to scale or shift the analog output range?
4. How can errors in resistor values affect the accuracy of a binary-weighted DAC?
5. How would you interface this DAC with a microcontroller in a practical application?

Post lab Viva Questions

1. How would you implement a DAC using resistors and an op-amp (IC 741)?
2. How does the binary-weighted resistor DAC work, and how would you choose resistor values?
3. How do you apply Kirchhoff's Current Law (KCL) in analyzing a DAC circuit?
4. How would you simulate the DAC circuit in Multisim or any other simulator?
5. What role does the op-amp play in improving the output of a DAC?

Result

The 4-bit binary-weighted resistor DAC was successfully simulated in NI Multisim. The output analog voltage corresponded accurately to the 4-bit digital inputs as verified using the theoretical values.

Ex No:9

**DESIGN AND SIMULATION OF
MULTIVIBRATORS**

Date:

Objective

To simulate Astable and Monostable multivibrators using IC 555 timer in NI Multisim and analyze their waveforms.

Software Required

NI Multisim Software 14.3

Theory

Multivibrators are electronic circuits used to implement various simple two-state devices such as oscillators, timers, and flip-flops. The IC 555 timer can be configured in different modes to function as astable or monostable multivibrators.

IC 555 Timer Overview

The IC 555 timer is a highly stable and versatile integrated circuit used for generating accurate time delays, oscillations, and pulses. It can operate in three modes: monostable, astable, and bistable. Internally, the 555 timer consists of two voltage comparators, an SR flip-flop, a discharge transistor, and a voltage divider network.

Pin functions:

- Pin 2 – Trigger
- Pin 3 – Output
- Pin 5 – Control Voltage
- Pin 6 – Threshold
- Pin 7 – Discharge

It operates reliably across a supply voltage range of 4.5 V to 15 V and is commonly used in waveform generation, PWM, and timing circuits.

Working Principle**1. Astable Multivibrator**

- No stable state – output continuously toggles between HIGH and LOW.
- The capacitor charges and discharges through resistors R_1 and R_2 .
- Generates a continuous square wave.

Time period:

$$T = 0.693 \times (R_1 + 2R_2) \times C$$

Theory – Continued

Frequency:

$$f = \frac{1.44}{(R_1 + 2R_2)C}$$

2. Monostable Multivibrator

- One stable state and one quasi-stable state.
- Triggered by a negative pulse on pin 2.
- Output goes HIGH temporarily, then returns LOW.

Pulse width:

$$T = 1.1 \times R \times C$$

Implementation in Simulation

- **Astable Mode:**
 - Connect resistors R_1 and R_2 between V_{cc} , pin 7, and pin 6.
 - Connect a capacitor from pins 6 and 2 to ground.
 - Output waveform is observed at pin 3.
- **Monostable Mode:**
 - Use a resistor and capacitor to determine the pulse width.
 - Apply a negative trigger pulse to pin 2.
 - Output pulse appears at pin 3.

Key Features

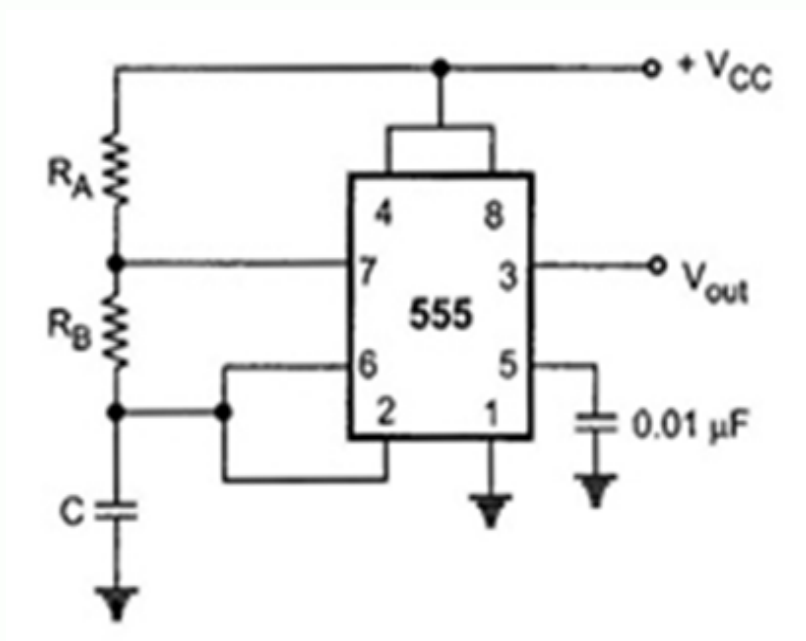
- Simple multivibrator design using IC 555
- Continuous square wave output in astable mode
- Single-shot output in monostable mode
- Output characteristics depend on external resistor and capacitor values

Applications

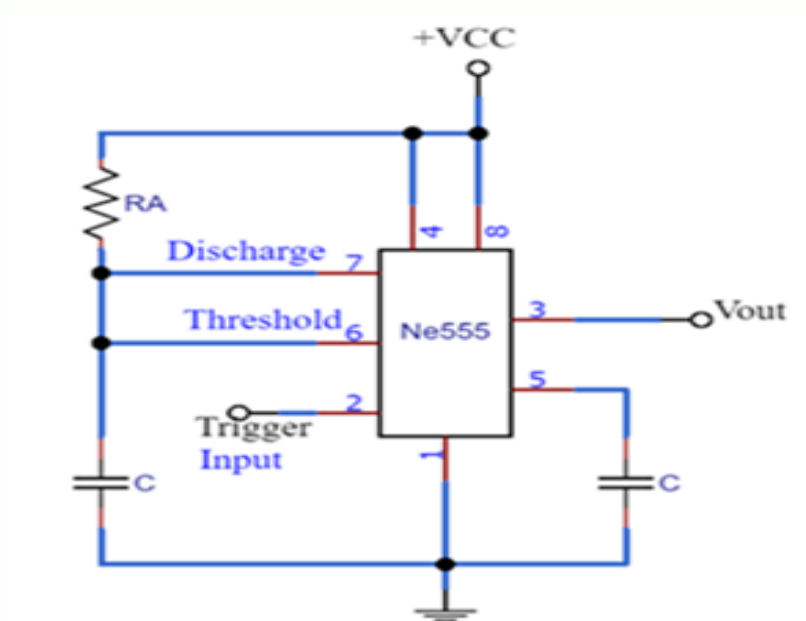
- Pulse generation
- Timer circuits
- Frequency generators
- Digital clock signals
- LED and relay drivers

Circuit Diagram

ASTABLE MULTIVIBRATOR USING 555 TIMER



MONOSTABLE MULTIVIBRATOR USING 555 TIMER



Design Calculation

ASTABLE MULTIVIBRATOR USING 555 TIMER

- Desired frequency = 6 kHz
- Time period

$$T = \frac{1}{f} = \frac{1}{6000} = 0.0001667 \text{ s}$$

- Using

$$T = 0.693 \times (R1 + 2R2) \times C$$

- Choose $C = 0.01 \mu F = 10^{-8} F$
- Rearranging:

$$R1 + 2R2 = \frac{T}{0.693 \times C} = \frac{0.0001667}{0.693 \times 10^{-8}} \approx 24.05 \text{ k}\Omega$$

- Choose: $R1 = 4.7 \text{ k}\Omega$, $R2 = 9.75 \text{ k}\Omega$ (standard $\approx 10 \text{ k}\Omega$)
- Final frequency:

$$f = \frac{1.44}{(4.7k + 2 \times 10k) \times 0.01 \times 10^{-6}} \approx 6 \text{ kHz}$$

MONOSTABLE MULTIVIBRATOR USING 555 TIMER

- Desired pulse width = 1.1 ms = 0.0011 s
- Using

$$T = 1.1 \times R \times C$$

- Choose $R = 10 \text{ k}\Omega$
- Calculate C :

$$C = \frac{T}{1.1 \times R} = \frac{0.0011}{1.1 \times 10,000} = 0.0000001 \text{ F} = 0.1 \mu F$$

- Choose: $R = 10 \text{ k}\Omega$, $C = 0.1 \mu F$

Procedure

ASTABLE MULTIVIBRATOR USING 555 TIMER

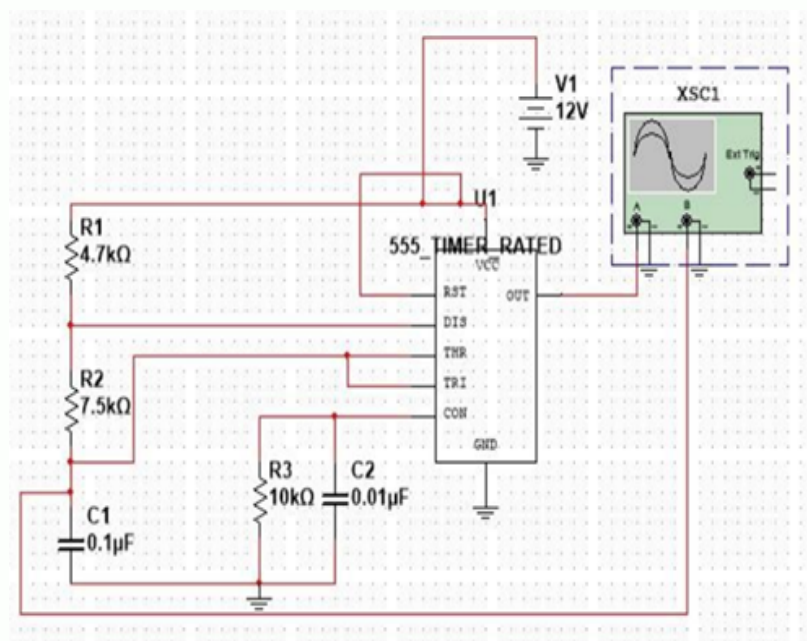
1. Open **NI Multisim** and start a new schematic.
2. Connect the **555 timer** in astable mode as per the circuit diagram.
3. Set the component values as follows:
 - $R_1 = 4.7\text{ k}\Omega$
 - $R_2 = 10\text{ k}\Omega$
 - $C = 0.01\text{ }\mu\text{F}$
4. Power the circuit with +5 V or +12 V as required.
5. Observe the output waveform on **CRO** or **oscilloscope**.
6. Measure the **frequency** and **amplitude** of the output waveform. Compare the measured values with the theoretical calculations.

MONOSTABLE MULTIVIBRATOR USING 555 TIMER

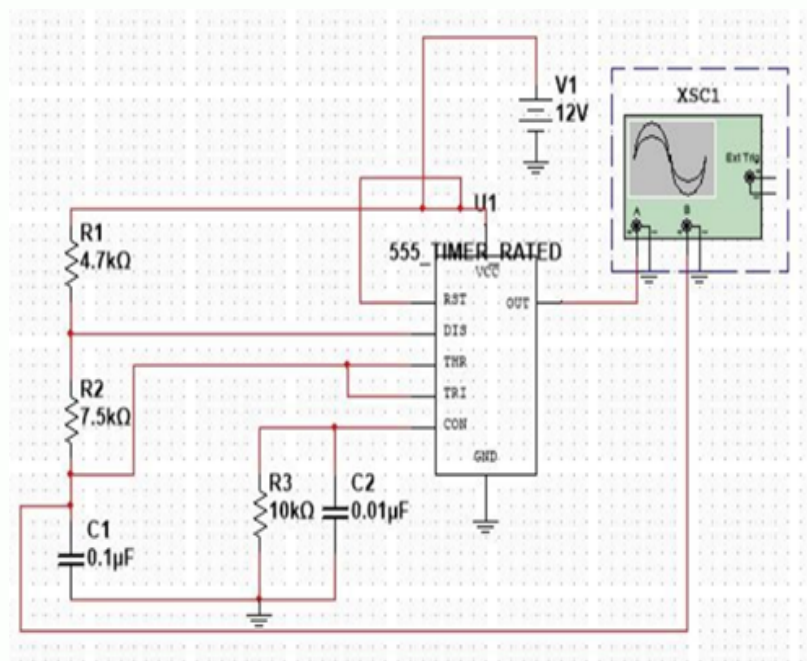
1. Open **NI Multisim** and start a new schematic.
2. Connect the **555 timer** in astable mode as per the circuit diagram.
3. Set the component values as follows:
 - $R_1 = 4.7\text{ k}\Omega$
 - $R_2 = 10\text{ k}\Omega$
 - $C = 0.01\text{ }\mu\text{F}$
4. Power the circuit with either +5 V or +12 V, as required.
5. Observe the output waveform using a **CRO** or **oscilloscope**.
6. Measure the **frequency** and **amplitude** of the output waveform and compare the results with the theoretical values.

Circuit Schematic

ASTABLE MULTIVIBRATOR USING 555 TIMER

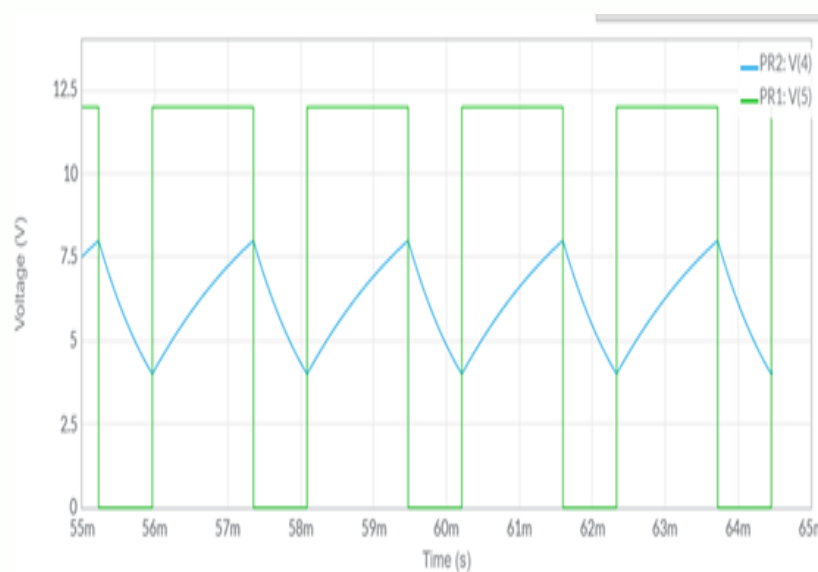
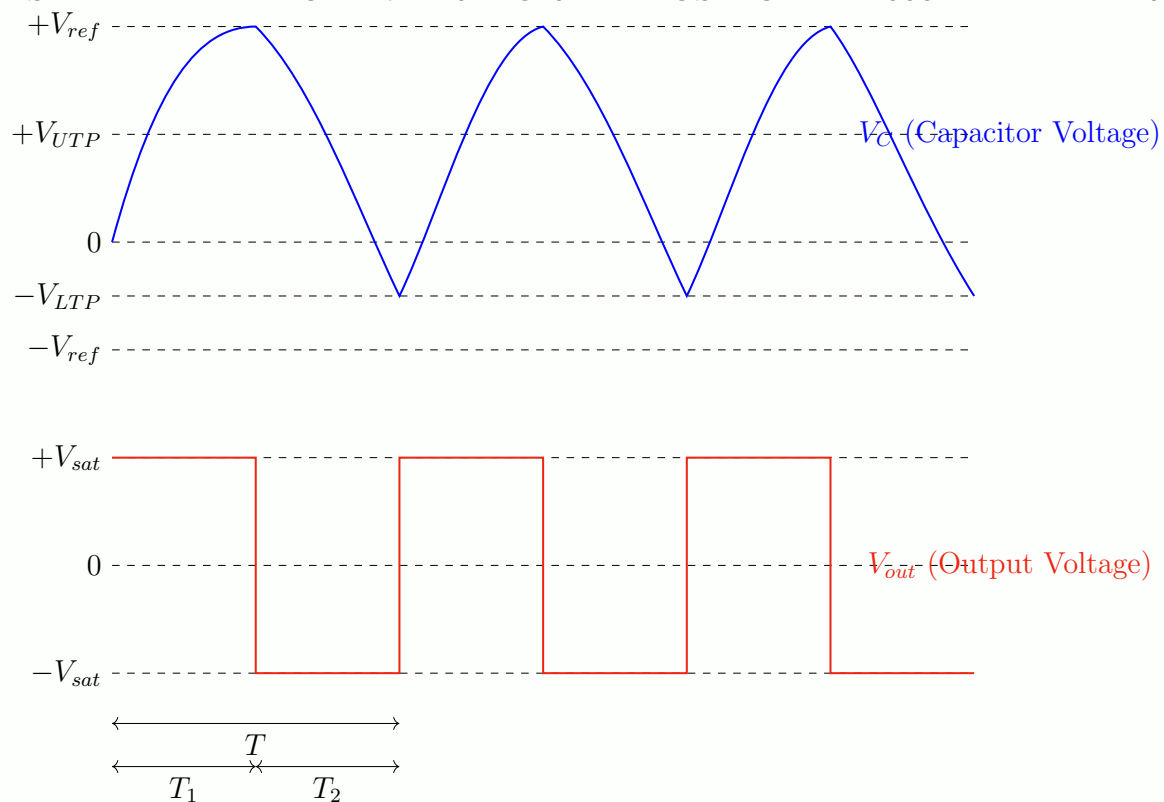


MONOSTABLE MULTIVIBRATOR USING 555 TIMER



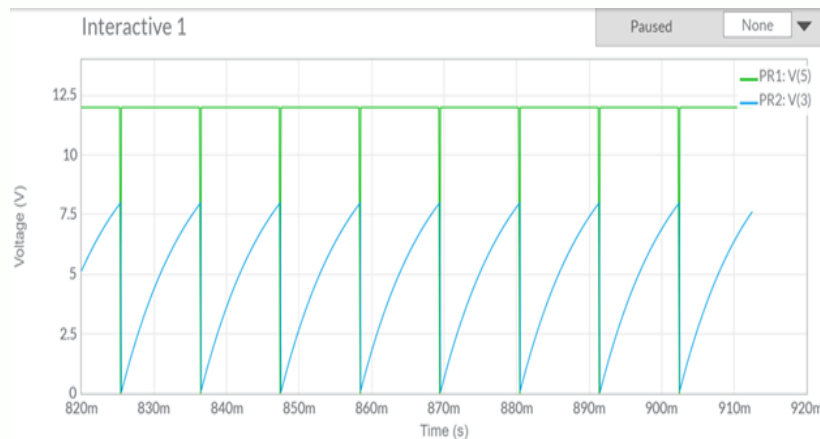
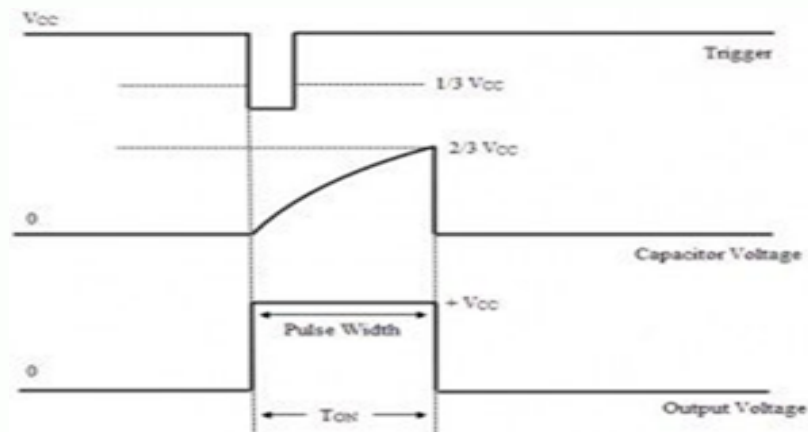
Model Graph

ASTABLE MULTIVIBRATOR USING 555 TIMER



Model Graph– Continued

MONOSTABLE MULTIVIBRATOR USING 555 TIMER



Pre lab Viva Questions

1. How would you configure a 555 timer in astable and monostable modes?
2. How can an op-amp be used to design an astable multivibrator circuit?
3. How do the values of resistors and capacitors affect the timing of multivibrator circuits?
4. How would you simulate triggering of a monostable multivibrator in Multisim?
5. How can you differentiate the waveform outputs of astable and monostable circuits?

Post lab Viva Questions

1. How did the output waveform change with variations in resistor and capacitor values during simulation?
2. How did you measure the pulse width and frequency from the simulated waveform?
3. How would you correct errors if the monostable multivibrator did not retrigger properly?
4. How do op-amp-based multivibrators compare with 555 timer-based circuits in per-

Post lab Viva Questions – Continued

formance?

5. How can these circuits be applied in timing, waveform generation, or digital triggering systems?

Result

The Astable and Monostable multivibrators were simulated using the IC 555 timer. The output waveforms were observed and compared with theoretical and practical results.

Ex No:10	SIMULATION OF WIEN BRIDGE AND RC PHASE SHIFT OSCILLATORS
Date:	

Objective

To simulate Wien Bridge Oscillator and RC Phase Shift Oscillator using NI Multisim and observe their output waveforms and frequency characteristics.

Software Required

NI Multisim Software 14.3

Theory

Wien Bridge Oscillator:

- Used for generating low-distortion sine waves in the **audio frequency range (AF)**.
- Comprises a **series RC** and a **parallel RC** network in a bridge configuration.
- Oscillation condition: **Total phase shift** = 0° and loop gain $A_v \times \beta = 1$.
- Frequency of oscillation is given by:

$$f_0 = \frac{1}{2\pi RC}$$

- For sustained oscillations:

$$\beta = \frac{1}{3}, \quad \text{so} \quad A_v = 3$$

- Slightly increasing gain above 3 ensures oscillations without distortion.

Key Features:

- Produces stable, low-distortion sine waves.
- Frequency determined by resistor-capacitor combinations.
- Easy amplitude control using automatic gain adjustment.
- Suitable for audio frequency applications.

Applications:

- Sine wave generators
- Function generators
- Audio signal testing
- Oscillator circuits in **analog** equipment

RC Phase Shift Oscillator:

- Consists of an **inverting amplifier** and a **three-section RC feedback network**.
- Each RC section provides a 60° phase shift, totaling 180° , combined with 180° from

Theory – Continued

the amplifier to satisfy the **Barkhausen criterion**.

- Frequency of oscillation:

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

- Commonly used for **low-frequency sine wave generation**.

Key Features:

- Simple design using passive components.
- Produces sine wave output without external triggering.
- Frequency is set by identical RC values.
- Suitable for low-frequency applications.

Applications:

- Signal generators
- Audio oscillators
- Laboratory test equipment
- Sine wave reference circuits

Design Calculation

RC Phase Shift Oscillator

- Desired frequency: $f = 2 \text{ kHz}$
- Chosen capacitance: $C = 0.1 \text{ }\mu\text{F}$
- Required resistance is calculated as:

$$R = \frac{1}{2\pi fC\sqrt{6}} = 330 \text{ }\Omega$$

Wien Bridge Oscillator

- Desired frequency: $f_0 = 20 \text{ kHz}$
- Chosen capacitance: $C = 0.01 \text{ }\mu\text{F}$
- Required resistance is calculated as:

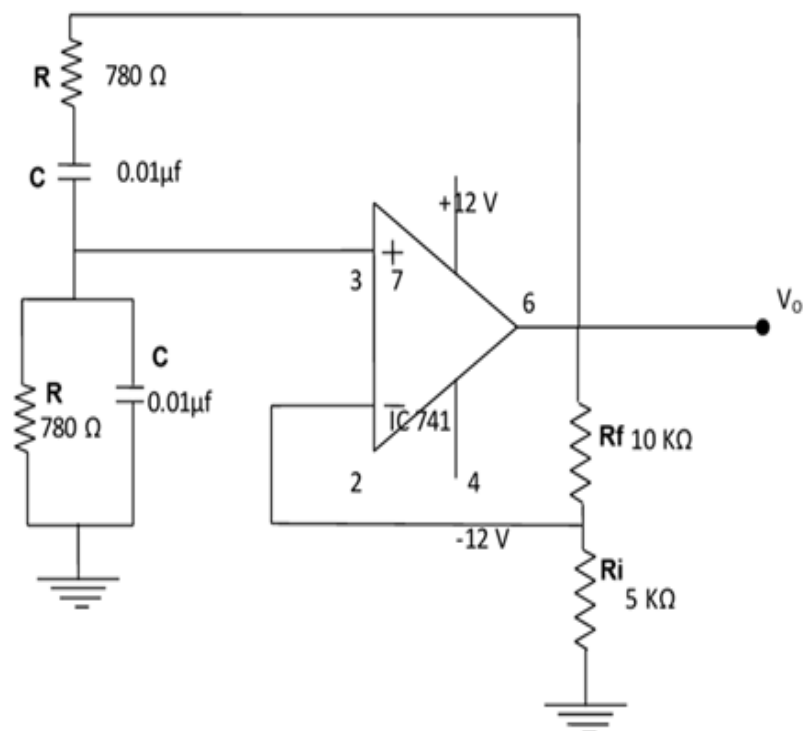
$$R = \frac{1}{2\pi f_0 C} = 780 \text{ }\Omega$$

- Assume $R_1 = 5 \text{ k}\Omega$, then feedback resistor:

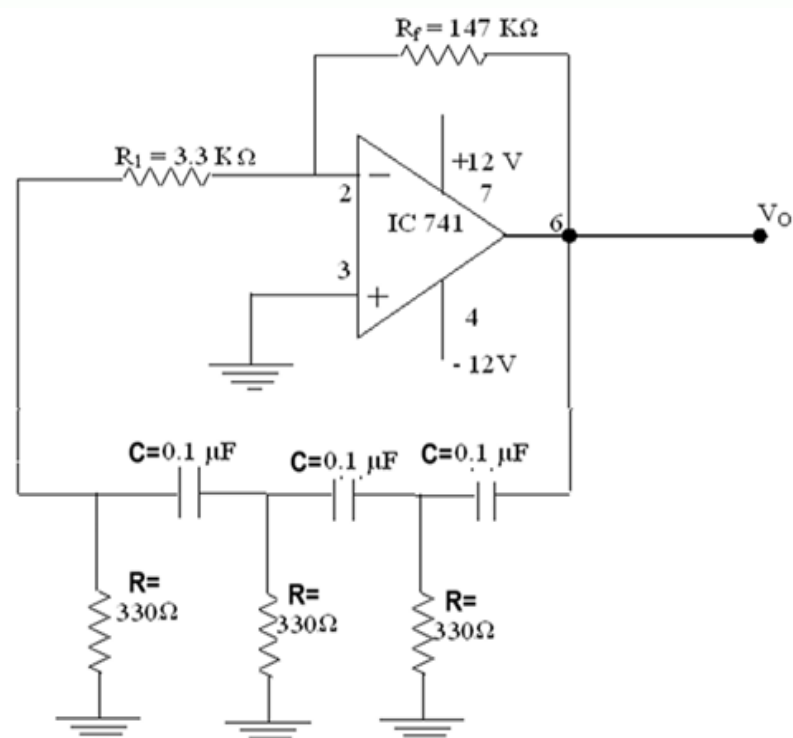
$$R_f = 2R_1 = 10 \text{ k}\Omega$$

Circuit Diagram

Wien Bridge Oscillator:

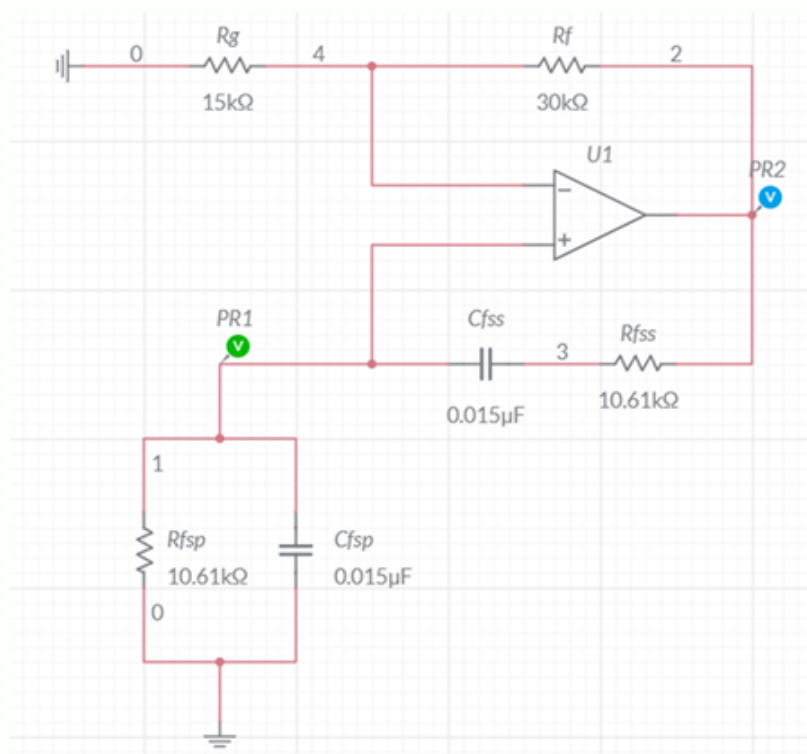


RC Phase Shift Oscillator

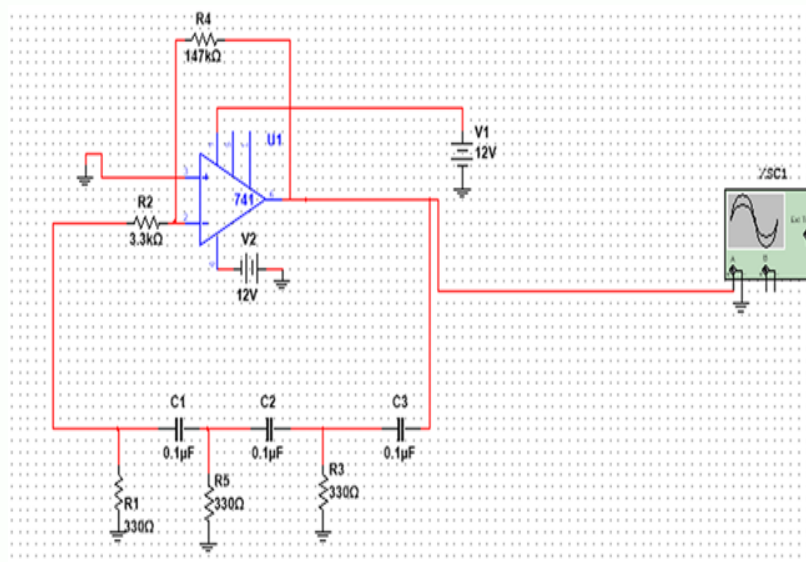


Circuit Schematic

Wien Bridge Oscillator:



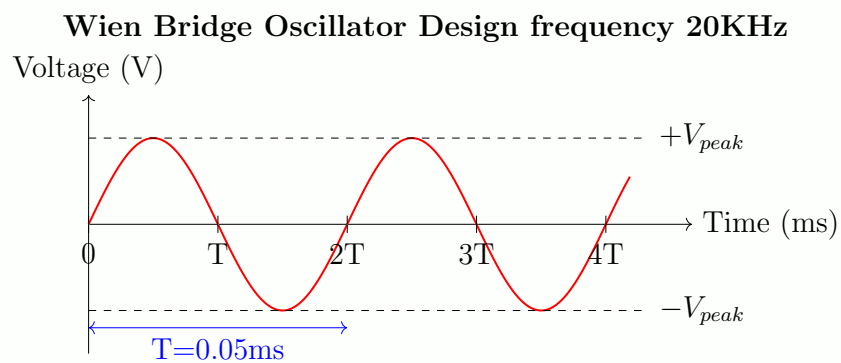
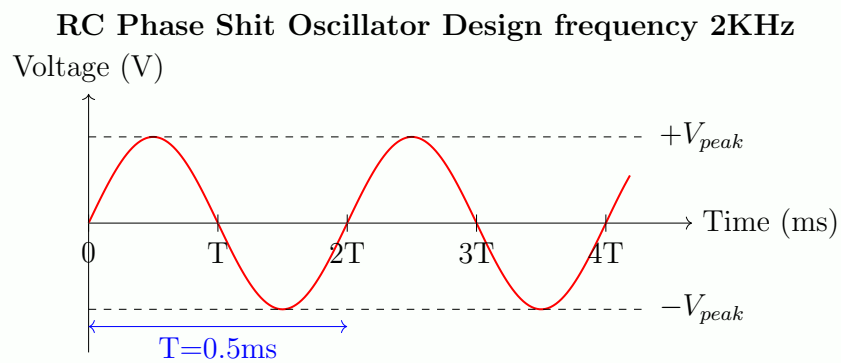
RC Phase Shift Oscillator



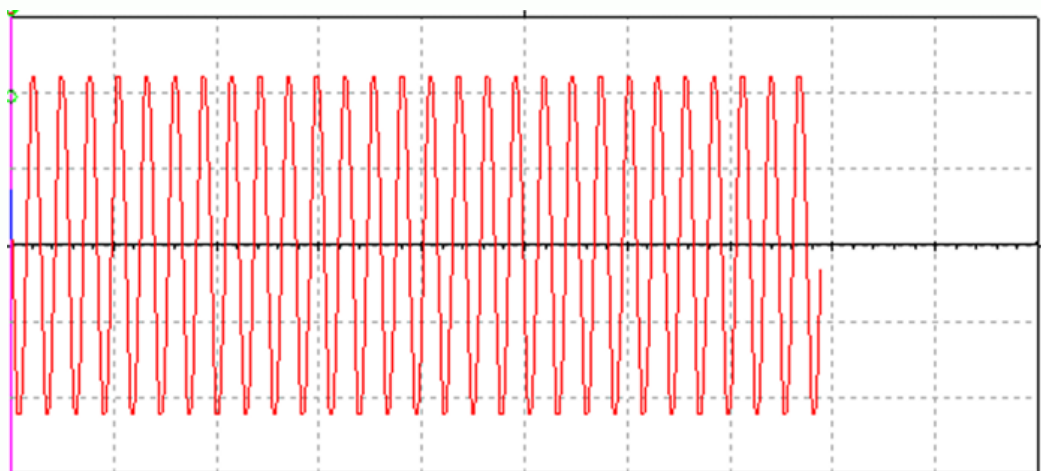
Procedure :

1. Open **NI Multisim** and create a new schematic.
2. Design and connect the **Wien Bridge Oscillator** and **RC Phase Shift Oscillator** circuits based on the calculated component values.
3. Power the circuit with $\pm 12\text{ V}$ (if using Op-Amp IC 741).
4. Run the simulation and observe the output waveform using the oscilloscope tool.
5. Measure the frequency of oscillation and compare it with the theoretical value to verify accuracy.

Model Graph



Sample Output Waveform



Pre lab Viva Questions

RC Phase shift Oscillator

1. What is the principle behind an RC phase shift oscillator?
2. Why are three RC sections used in the feedback path of an RC phase shift oscillator?
3. What condition must be satisfied for sustained oscillations?
4. What is the role of the op-amp in the oscillator circuit?
5. Derive the expression for the frequency of oscillation of an RC phase shift oscillator.

Wien Bridge Oscillator

1. What is the purpose of a Wien bridge oscillator?
2. Why do we use a combination of series and parallel RC in the feedback network?
3. State the expression for the frequency of oscillation in a Wien bridge oscillator.
4. Why is a gain control mechanism important in a Wien bridge oscillator?
5. What is the Barkhausen criterion?

Post lab Viva Questions

RC Phase Shift Oscillator

1. If the frequency of oscillation is lower than expected, which component value (R or C) should be adjusted? Justify.
2. How would the waveform be affected if one capacitor in the RC network is shorted?
3. Suppose you replace the op-amp with one having lower gain. What would be the impact on oscillation?
4. How would you design the circuit to operate at 1 kHz? Explain with component values.
5. If the output waveform is distorted, what probable circuit faults would you check and why?

Wien Bridge Oscillator

1. How would you choose R and C to generate a 10 kHz sine wave?
2. If the gain in the op-amp circuit is slightly less than 3, what will happen to the output? Explain.
3. How does the circuit maintain constant amplitude in the output waveform?
4. If you observe clipping in the output waveform, what changes would you suggest in the circuit?
5. Design a Wien bridge oscillator for 2 kHz and explain the selection of each component.

Result

The Wien Bridge and RC Phase Shift Oscillators were simulated using NI Multisim. The output waveforms were observed and the measured frequencies closely matched theoretical calculations, demonstrating proper oscillator operation.